

Untitled hardware project

Class haps	Feasibility feasible	BoM Readiness Ready	Score 7.7
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Executive Summary

Metric	Value	Unit	Confidence	Notes
Station-keeping endurance	30	days	medium	Mission endurance requested in the brief before solar energy balance, battery night-cycle and stratospheric wind validation.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_ready_for_architecture

6/6 intake areas pass; 0 review, 0 blocked. Class haps (high); requirements 3; numeric facts 3; requirement trace 3/3 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	351 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	pass	1	3 quantified requirement(s), 3 numeric fact(s).	none	Quantified requirements are sufficient for a first architecture pass.
product_class_selection	pass	1	Selected haps with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	haps has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	3 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

no_clarification_needed

No clarification questions are needed before scratch architecture generation.

Architecture Admission Gate

architecture_generation_admitted

6/6 admission areas pass; 0 review, 0 blocked. Can use for review: yes; can proceed to BoM: yes. Source scratch_universal_architecture; next action: none.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	pass	1	6/6 intake areas pass; class haps (high); 3 quantified requirement(s).	none	Brief intake is strong enough to admit scratch architecture generation.
clarification_clearance	pass	1	0 clarification question(s): 0 required, 0 recommended, 0 architecture-blocking.	none	No brief clarification is needed before architecture generation.
class_grammar_clearance	pass	1	Product class haps; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 11; submodules 33; components 132.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 8/8.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_hold	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class haps; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 294; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class haps; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	565 provenance ref(s); 294 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	132 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufacturer/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifier lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

2/6 freeze areas pass; 4 review, 0 blocked. Modules/submodules/components 11/33/132. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Treat the architecture as review-ready until module review questions and critical sourcing blocks close..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	pass	1	Architecture admission architecture_generation_admitted; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Architecture was admitted from the scratch flow with clean stage and lineage evidence.
module_structure	review	0.75	Component allocation allocation_ready; module engineering module_engineering_review_required; 0/11 modules pass; 9 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 8/8 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 3/3 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.67	Pre-BoM engineering engineering_review_ready; 4/6 areas pass; calculations needing review 2; assumptions needing review 23.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	132 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 48 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

4 closure row(s): 4 ready, 0 blocked. Queues S/E/V/R 1/1/2/0; next row freeze-close:1:module_structure. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
2	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
3	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv
4	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.verification-plan.csv

component_candidates_ready_for_sourcing

7/7 candidate areas pass; 0 review, 0 blocked. Candidate lines 132; critical 12/12; worklist rows 132; duplicate groups 0; provenance violations 0; ready for sourcing yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	132 candidate BoM line(s), 132 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	pass	1	132/132 candidate line(s) have non-generic identity, description and role context.	none	Candidate identities are specific enough for first-pass sourcing worklists.
quantity_basis	pass	1	132/132 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	12/12 class-critical part(s) are represented by candidate lines; 12 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	132/132 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorized

6/6 sourcing-authorization areas pass; 0 review, 0 blocked. Critical intake 12/12; full intake 132/132; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorization	pass	1	architecture_generation_admitted; review yes, BoM yes.	none	Architecture admission permits evidence-gated sourcing work.
component_candidate_authorization	pass	1	component_candidates_ready_for_sourcing; 132 candidate line(s), 132 sourcing-worklist row(s), 0 provenance violation(s).	none	Component candidates are concrete enough for sourcing intake.
critical_intake_scope	pass	1	12/12 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	132/132 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_authorized

Display mode candidate_only. 2/7 BoM-admission areas pass; 5 review, 0 blocked. Critical priced 0/12; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	pass	1	sourcing_authorized; authorized yes; 6/6 areas pass.	none	Sourcing is authorized, so the BoM may be shown as a candidate sourcing queue.
candidate_bom_scope	pass	1	132 candidate BoM line(s), 12 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/132 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/12 critical line(s) have admitted unit costs; 12 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.93	0 source-backed claim(s), 48 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 12. Verification accepted: 0/35; unaccepted: 35. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 12 critical BoM line(s) lack source-backed cost evidence.
- 9 verification activity/activities remain blocked.
- 0/35 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	12/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

4/6 areas pass; 2 review, 0 blocked. Modules/submodules/components: 11/33/132. Calculations within envelope 1/3; calculation blockers 0. Assumption review rows 23; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	11 modules, 33 submodules and 132 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	33/33 submodules ready; 12/12 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	8/8 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	3/3 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	review	0.33	1/3 calculations are within envelope; 2 need review; 0 block.	none	Attach reviewer evidence to calculations marked needs_review.
engineering_assumptions	review	0.32	11/34 assumptions are brief-supported or model-present; 23 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

1/7 costing areas pass; 0 review, 6 blocked. Critical priced lines 0/12; unpriced critical lines 12. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/132 BoM lines have admitted pricing; rejected source records 0.	structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_harvesting_solar_skin_array_solar_cell_array: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_harvesting_mppt_conversion_mppt_power_tracker: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_stratospheric_battery_pack_stratospheric_battery_pack: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_electric_propulsion_pods_electric_propulsion_pod: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 7 more	Admit source-backed supplier, manufacturer, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/12 critical lines have admitted unit costs.	structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure: High-aspect-ratio wing structure energy_harvesting_solar_skin_array_solar_cell_array: Solar cell array energy_harvesting_mppt_conversion_mppt_power_tracker: MPPT power tracker energy_storage_source_stratospheric_battery_pack_stratospheric_battery_pack	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				k: Stratospheric battery pack actuation_kinematics_electric_propulsion_pods_electric_propulsion_pod: Electric propulsion pod plus 7 more	
source_backed_provenance	blocked	0.93	0 source-backed claims; 48 critical missing-source claims; 0 provenance violations.	structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/manufacture: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_harvesting_solar_skin_array_solar_cell_array/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 43 more	Keep supplier, manufacturer, MPN, unit cost and lead-time fields empty until source-backed evidence is admitted.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line-level costing.
sourcing_evidence_authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.
source_reference_quality	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 0 fresh timestamp.	none	Add source references before costing review.
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	12 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

52 missing-evidence row(s): 15 sourcing, 37 verification. Ready 49; blocked 0. Blocks BoM 15; blocks publish 52. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	high_aspect_ratio_wing_structure	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	solar_cell_array	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_intake	mppt_power_tracker	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
9	sourcing	blocker	ready_for_intake	stratospheric_battery_pack	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
10	sourcing	blocker	ready_for_intake	electric_propulsion_pod	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_in take	flight_control_compute r	supplier_catalogue_or_quote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	sourcing	blocker	ready_for_in take	ground_control_link	supplier_catalogue_or_quote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
13	sourcing	blocker	ready_for_in take	gnss_ins_navigation_unit	supplier_catalogue_or_quote	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						ote; retrievedAt		
14	sourcing	blocker	ready_for_in take	stratospheri c_communi cations_pay load	supplier_cat alogue_or_q uote	component WordId; supplierNa me; manufactur er; mpn where applicable; unitCostGb p; leadTimeWe eks where stated; sourceGrad e; evidence.ref ; evidence.qu ote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
15	sourcing	blocker	ready_for_in take	payload_ant enna_array	supplier_cat alogue_or_q uote	component WordId; supplierNa me; manufactur er; mpn where applicable; unitCostGb p; leadTimeWe eks where stated; sourceGrad e; evidence.ref ; evidence.qu ote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
16	sourcing	blocker	ready_for_intake	thermal_insulation_blanket	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
17	sourcing	blocker	ready_for_intake	recovery_parachute_system	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
18	verification	major	ready_for_intake	design_review:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	calculation:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef;	anonymous reviewer; unchecked model assertion; test-fixture://	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidenceNote; reviewedAt	review; source-evidence claims that belong in sourcing intake	
20	verification	major	ready_for_intake	design_review:energy_harvesting	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
21	verification	major	ready_for_intake	calculation:energy_harvesting	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
22	verification	major	ready_for_intake	design_review:energy_storage_source	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
23	verification	major	ready_for_intake	calculation:energy_storage_source	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
							sourcing intake	
24	verification	major	ready_for_intake	design_review:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	calculation: power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
26	verification	major	ready_for_intake	design_review:actuation_kinematics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
27	verification	major	ready_for_intake	calculation: actuation_kinematics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
28	verification	major	ready_for_intake	design_review:control_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
29	verification	major	ready_for_intake	calculation: control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
30	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
31	verification	major	ready_for_intake	calculation: sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
32	verification	major	ready_for_intake	design_review:payload_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
33	verification	major	ready_for_intake	design_review:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
34	verification	major	ready_for_intake	calculation:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
35	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
36	verification	major	ready_for_intake	calculation:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
37	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
38	verification	major	ready_for_intake	calculation:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
39	verification	major	ready_for_intake	interface_review:energy_harvesting:power_distribution:solar_dc_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
40	verification	major	ready_for_intake	interface_review:energy_storage_source:power_distribution:battery_dc_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
41	verification	major	ready_for_in take	interface_re view:power_ distribution: actuation_ki nematics:pr opulsion_po wer_bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
42	verification	major	ready_for_in take	interface_re view:control _compute_c ommunicati on:actuation _kinematics :flight_contr ol_bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
43	verification	major	ready_for_in take	interface_re view:sensin g_instrume ntation:cont rol_comput e_communi cation:sens or_bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
44	verification	major	ready_for_intake	interface_review:payload_communication:control_compute_communication:payload_data_link	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
45	verification	major	ready_for_intake	interface_review:environmental_interface:energy_storage_source:thermal_path	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
46	verification	major	ready_for_intake	interface_review:safety_protection:control_compute_communication:telemetry_link	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
47	verification	major	ready_for_intake	compliance_review:CAA-UAS-OPS	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
48	verification	major	ready_for_intake	compliance_review:DO-160-ENV	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
49	verification	major	ready_for_intake	compliance_review:SO LAR- ENERGY- BALANCE- REVIEW	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
50	verification	major	ready_for_intake	risk_review:1	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
51	verification	major	ready_for_intake	risk_review:2	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
52	verification	major	ready_for_intake	risk_review:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/35. Source-backed BoM claims 0; critical missing BoM claims 48; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 44 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 4 review-required claim(s).
- bom sourcing has 60 blocked/source-required claim(s) and 120 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).
- structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_harvesting_solar_skin_array_solar_cell_array/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_harvesting_solar_skin_array_solar_cell_array/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_harvesting_solar_skin_array_solar_cell_array/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	11 modules, 33 submodules and 8 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0.01	3/239 claims pass; 62 blocked/source-required, 174 review-required.	architecture design has 0 blocked/source-required claim(s) and 44 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 4 review-required claim(s). bom sourcing has 60 blocked/source-required claim(s) and 120 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/3 requirements have accepted assurance; 3 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/35 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 35 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.93	0 source-backed BoM claims, 48 critical missing-source claims, 0 provenance violations.	structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_wing_primary_structure_high_aspe	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				ct_ratio_wing_structure/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_harvesting_solar_skin_array_solar_cell_array/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 43 more	

evidence_authenticity	review	0	0/0 evidence row(s) are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.
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Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
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Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 64; trust blockers 61. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	15	52	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	11	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	37	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance -domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

239 claims: 3 brief-supplied, 0 reviewer-accepted, 0 source-backed, 4 calculated needing review, 170 generated needing review, 62 source-required, 0 blocked. Blocking: 62 BoM, 235 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Electric propulsion pod / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Electric propulsion pod / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Electric propulsion pod / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Electric propulsion pod / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight control computer / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight control computer / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Flight control computer / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Flight control computer / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Ground control link / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Ground control link / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Ground control link / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Ground control link / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	MPPT power tracker / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	MPPT power tracker / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	MPPT power tracker / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	MPPT power tracker / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Solar cell array / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Solar cell array / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Solar cell array / supplier	supplier is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Solar cell array / unit_cost_g bp	unit_cost_g bp is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Stratospheri c battery pack / manufactur er	manufactur er is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Stratospheri c battery pack / mpn	mpn is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Stratospheri c battery pack / supplier	supplier is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Stratospheri c battery	unit_cost_g bp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			pack / unit_cost_g bp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thermal insulation blanket / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thermal insulation blanket / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thermal insulation blanket / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thermal insulation blanket / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Payload antenna array / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Payload antenna array / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Payload antenna array / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Payload antenna array / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Stratospheric communications payload / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Stratospheric communications payload / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Stratospheric communications payload / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Stratospheric communications payload / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Recovery parachute system / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Recovery parachute system / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Recovery parachute system / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Recovery parachute system / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	GNSS INS navigation unit / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	GNSS INS navigation unit / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	GNSS INS navigation unit / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	GNSS INS navigation unit / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	High-aspect-ratio wing structure / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	High-aspect-ratio wing structure / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	High-aspect-ratio wing	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			structure / supplier					nufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	High-aspect-ratio wing structure / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
component_candidate	source_required	generated_model	Electric propulsion pod	Electric propulsion pod is an architecture-derived component candidate with quantity 4 each.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Flight control computer	Flight control computer is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Ground control link	Ground control link is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	MPPT power tracker	MPPT power tracker is an architecture	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier,

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	Solar cell array	-derived component candidate with quantity 1 candidate.				manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
				Solar cell array is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Stratospheric battery pack	Stratospheric battery pack is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Thermal insulation blanket	Thermal insulation blanket is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	Payload antenna array	Payload antenna array is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Stratospheric communications payload	Stratospheric communications payload is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Recovery parachute system	Recovery parachute system is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	GNSS INS navigation unit	GNSS INS navigation unit is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	High-aspect-ratio wing structure	High-aspect-ratio wing structure is an architecture-derived component	model:scratch_universal_architecture.haps	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				candidate with quantity 1 candidate.				time evidence through sourcing intake.
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:haps.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
component_candidate	generated_needs_review	generated_model	control linkage rod set	control linkage rod set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	low-temperature servo actuator	low-temperature servo actuator is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	servo heater trace	servo heater trace is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	surface position sensor	surface position sensor is an architecture-derived component candidate	model:scratch_universal_architecture.haps	no	yes	Review whether this is a unique physical item, shared component or separate

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				with quantity 1 candidate.				installation instance.
component_candidate	generated_need_review	generated_model	high-altitude propeller set	high-altitude propeller set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	motor controller module	motor controller module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	propulsion pod fairing	propulsion pod fairing is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	airspeed setpoint manager	airspeed setpoint manager is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.haps	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 239 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

3/239 claims pass by being brief-supplied, source-backed or reviewer-accepted. 62 blocked/source-required claims, 174 review-required claims. Claim pass ratio 0.01. Blocking: 62 BoM, 235 publishable.

- architecture design has 0 blocked/source-required claim(s) and 44 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 4 review-required claim(s).
- bom sourcing has 60 blocked/source-required claim(s) and 120 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 6 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	3	3	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	44	0	44	0	0	0	44	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	6	0	4	2	2	2	5	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	180	0	120	60	60	60	180	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	6	0	6	0	0	0	6	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

64 open evidence gaps: 18 blockers, 46 major, 0 minor. Closure paths: 15 sourcing intake, 37 verification intake, 11 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 15, publishable 64.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	12 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	9 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/35 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
calculation_review	major	needs review	HAPS endurance hours	720 h endurance target before stratospheric energy-balance proof.	engineering_review	publishable	Confirm solar resource, battery reserve, wind field, degradation, payload power and recovery plan.
calculation_review	major	needs review	HAPS wingspan-to-altitude marker	1.75 m/km is a trace marker, not an aerodynamic proof.	engineering_review	publishable	Confirm wing area, mass budget, airfoil, aeroelastic margin, launch loads and transport constraints.
assumption_review	major	needs review	Station-keeping endurance	Station-keeping endurance uses endurance_days with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	Stratospheric Altitude Band	Stratospheric Altitude Band currently evaluates as pass against 15 to 25 km for common HAPS station-keeping concepts; outside this band needs atmospheric,	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				power and airspace evidence.			
assumption_review	major	needs review	Station-Keeping Endurance	Station-Keeping Endurance currently evaluates as pass against <=60 days plausible for early solar HAPS concept, 60-180 days needs strong degradation and energy evidence, >180 days is high risk.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Wing Span Envelope	Wing Span Envelope currently evaluates as pass against 10 to 80 m for lightweight demonstrator-to-service HAPS concepts before aeroelastic proof.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Solar/Energy/F light Closure	Solar/Energy/F light Closure currently evaluates as pass against solar DC, battery DC, propulsion power, flight control and battery monitor paths present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Payload/Telemetry/Safety Closure	Payload/Telemetry/Safety Closure currently evaluates as pass against payload data, payload power, telemetry, hardwired trip, thermal path and	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				recovery parachute present.			
assumption_review	major	needs review	CAA-UAS-OPS	High-altitude unmanned aircraft operational safety case is relevant to this product class in UK / aviation authority review.	engineering_review	publishable	Airspace concept of operations, detect-and-avoid assumptions, command link and recovery evidence.
assumption_review	major	needs review	DO-160-ENV	Airborne equipment environmental conditions is relevant to this product class in Aviation engineering reference.	engineering_review	publishable	Temperature, vibration, electrical power and EMC environmental evidence tailored to stratospheric operation.
assumption_review	major	needs review	SOLAR-ENERGY-BALANCE-REVIEW	Solar day/night energy balance review is relevant to this product class in Project engineering gate.	engineering_review	publishable	Solar collection, battery night-cycle, propulsion load and station-keeping wind model evidence.
sourcing_evidence	blocker	ready for intake	High-aspect-ratio wing structure	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Solar cell array	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	MPPT power tracker	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Stratospheric battery pack	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Electric propulsion pod	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Flight control computer	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	Ground control link	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	GNSS INS navigation unit	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Stratospheric communications payload	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Payload antenna array	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

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Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	Thermal insulation blanket	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Recovery parachute system	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
verification_evidence	major	ready for intake	Ultra-Light Wing Structure	Review Ultra-Light Wing Structure architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Ultra-Light Wing Structure	Verify Ultra-Light Wing Structure requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Solar Energy Harvesting	Review Solar Energy Harvesting architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Solar Energy Harvesting	Verify Solar Energy Harvesting requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Night-Cycle Energy Storage	Review Night-Cycle Energy Storage architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							intake template.
verification_evidence	major	ready for intake	Night-Cycle Energy Storage	Verify Night-Cycle Energy Storage requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Power Distribution	Review Flight Power Distribution architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Power Distribution	Verify Flight Power Distribution requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Propulsion And Control Surfaces	Review Propulsion And Control Surfaces architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Propulsion And Control Surfaces	Verify Propulsion And Control Surfaces requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Autonomy Control Communication	Review Autonomy Control Communication architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Autonomy Control Communication	Verify Autonomy Control Communication requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Navigation And Airdata Sensing	Review Navigation And Airdata Sensing	verification_intake	publishable	Collect reviewer evidence through the verification

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				architecture allocation			intake template.
verification_evidence	major	ready for intake	Navigation And Airdata Sensing	Verify Navigation And Airdata Sensing requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Communications Payload	Review Communications Payload architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Stratospheric Environmental Interface	Review Stratospheric Environmental Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Stratospheric Environmental Interface	Verify Stratospheric Environmental Interface requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Safety And Recovery	Review Flight Safety And Recovery architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Safety And Recovery	Verify Flight Safety And Recovery requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Ground Handling Serviceability	Review Ground Handling Serviceability architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Ground Handling Serviceability	Verify Ground Handling Serviceability requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							intake template.
verification_evidence	major	ready for intake	Solar Energy Harvesting -> Flight Power Distribution	Verify solar_dc_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Night-Cycle Energy Storage -> Flight Power Distribution	Verify battery_dc_buss interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Power Distribution -> Propulsion And Control Surfaces	Verify propulsion_power_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Autonomy Control Communication -> Propulsion And Control Surfaces	Verify flight_control_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Navigation And Airdata Sensing -> Autonomy Control Communication	Verify sensor_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Communications Payload -> Autonomy Control Communication	Verify payload_data_link interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Stratospheric Environmental Interface -> Night-Cycle Energy Storage	Verify thermal_path interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Flight Safety And Recovery -> Autonomy Control	Verify telemetry_link interface contract	verification_intake	publishable	Collect reviewer evidence through the verification

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
			Communication				intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for CAA-UAS-OPS	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for DO-160-ENV	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for SOLAR-ENERGY-BALANCE-REVIEW	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Energy-negative overnight operation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Wing structural failure in gust or launch handling	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Lost command link or uncontrolled descent	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 15, engineering 11, verification 37, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	15	15/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:electric_propulsion_pod; sourcing:flight_control_computer; sourcing:gnss_ins_navigation_unit; sourcing:ground_control_link; sourcing:high_aspect_ratio_wing_structure; plus 7 more
3	Complete Engineering Review	11	0/11/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:CAA-UAS-OPS; assumption:compliance:DO-160-ENV; assumption:compliance:SO-LAR-ENERGY-BALANCE-REVIEW; assumption:metric:headline_output; assumption:safety:haps_altitude_band; assumption:safety:haps_endurance_target; assumption:safety:haps_energy_flight_closure; assumption:safety:haps_payload_comms_safety; plus 3 more
4	Accept Reviewer Evidence	37	2/35/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close	Every non-source verification activity is accepted, or explicitly deferred and removed from	readiness:3; readiness:4; verification:actuation_kinematics; verification:calculation_control_compute_c

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
					before	publication	ommunication
					publication.	criteria.	; verification:ca
							lculation:ener
							gy_harvesting
							; verification:ca
							lculation:ener
							gy_storage_s
							ource; verification:ca
							lculation:envir
							onmental_inte
							rface; verification:ca
							lculation:main
							tenance_servi
							ceability; plus
							12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

5 meeting benchmark, 2 below benchmark, average comparable ratio 0.99.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	11 modules	11 modules	1	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	33 submodules	33 submodules	1	meets	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	132 components	132 components	1	meets	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	55 links	40 links	1.38	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.75 ratio	0	below	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	44 activities	26 activities	1.69	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a solar-electric high-altitude pseudo-satellite for 20 km altitude, 30 day station-keeping endurance and 35 m wingspan with wing structure, solar cell array, MPPT power tracker, stratospheric battery pack, electric propulsion pods, flight control computer, GNSS INS navigation, communications payload, thermal insulation and recovery parachute.

Requirement	Value	Unit
Wingspan	35	m
Endurance	30	days
Operating altitude	20	km

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Wingspan	covered	35 m	Ultra-Light Wing Structure / Wing primary structure; Solar Energy Harvesting / Solar skin array; Propulsion And Control Surfaces / Electric propulsion pods; Ground Handling Serviceability / Wing assembly service	sanity:haps_wingspan_envelope; sanity:haps_energy_flight_closure	Wingspan is traced through structure, solar area, actuation and ground-handling serviceability. Requirement has architecture coverage plus metric or engineering sanity coverage.
Endurance	covered	30 days	Solar Energy Harvesting / Solar skin array; Night-Cycle Energy Storage / Stratospheric battery pack; Flight Power Distribution / Solar and battery power management; Propulsion And Control Surfaces / Electric propulsion pods; Autonomy Control Communication / Flight control computer	metric:headline_output; sanity:haps_endurance_target; sanity:haps_energy_flight_closure	Endurance is traced through solar harvesting, storage, distribution, propulsion and autonomy reserve logic. Requirement has architecture coverage plus metric or engineering sanity coverage.
Operating altitude	covered	20 km	Stratospheric Environmental Interface / Thermal protection; Navigation And Airdata Sensing / Navigation suite; Autonomy Control Communication / Flight control computer; Flight Safety And Recovery / Abort and recovery	sanity:haps_altitude_band; sanity:haps_energy_flight_closure; sanity:haps_payload_comms_safety	Altitude is traced through environmental protection, sensing, autonomy and flight-safety closure. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

3 requirement row(s): 0 accepted with evidence, 3 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Operating altitude	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	environmental_interface; sensing_instrumentation; control_compute_communication ; safety_protection	design_review:control_compute_communication; calculation:control_compute_communication; design_review:sensing_instrumentation; calculation:sensing_instrumentation; design_review:environmental_interface; calculation:environmental_interface; design_review:safety_protection ; calculation:safety_protection	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
Endurance	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	energy_harvesting; energy_storage_source; power_distribution; actuation_kinematics; control_compute_communication	design_review:energy_harvesting; calculation:energy_harvesting; design_review:energy_storage_source; calculation:energy_storage_source; design_review:power_distribution; calculation:power_distribution; design_review:actuation_kinematics; calculation:actuation_kinematics; design_review:control_compute_communication; calculation:control_compute_communication	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
Wingspan	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	structure_containment; energy_harvesting; actuation_kinematics; maintenance_serviceability	design_review:structure_containment; calculation:structure_containment; design_review:energy_harvesting; calculation:energy_harvesting; design_review:a	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
					ctuation_kinematics; calculation:actuation_kinematics; design_review:maintenance_serviceability; calculation:maintenance_serviceability	

Engineering Assurance Matrix

3 requirements: 0 accepted, 0 ready, 3 needing review, 0 blocked, 0 unlinked. 3 have calculation links, 3 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Operating altitude	needs review	covered	Stratospheric Environment; Navigation And Airdata Sensing; Autonomy Control; Communication; Flight Safety And Recovery	haps_altitude_ft; haps_wingspan_to_altitude_ratio	assumption_resolution: component: control_compute_communication_flight_control_computer_flight_control_computer; assumption_resolution: component: control_compute_communication_ground_communication_link_ground_control_link; assumption_resolution: component: environmental_interface_thermal_protection_thermal_insulation_blanket; assumption_resolution: component: safety_protection_abort_recovery_recovery_parachute_system; assumption_resolution: component: sensing_instrumentation_navigation_suite_gnss_ins_navigation_unit; plus 28 more	pending; pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.
Endurance	needs review	covered	Solar Energy Harvesting; Night-Cycle Energy Storage; Flight Power Distribution; Propulsion And Control Surfaces; Autonomy	haps_endurance_hours	assumption_resolution: component: actuation_kinemematics_electrical_propulsion_pods_electrical_propulsion_pod; assumption_resolution:	pending; pending; pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
			Control Communication		component: control_com pute_communication_flight_control_computer_flight_control_computer; assumption_resolution: component: control_com pute_communication_ground_command_link_ground_control_link; assumption_resolution: component: energy_harvesting_mpp t_conversion_mppt_power_tracker; assumption_resolution: component: energy_harvesting_solar_skin_array_solar_cell_array; plus 37 more			

Wingspan	needs review	covered	Ultra-Light Wing Structure; Solar Energy Harvesting; Propulsion And Control Surfaces; Ground Handling Serviceability	haps_wingspan_to_altitude_ratio	assumption_resolution: component: actuation_kinematics_electrical_propulsion_pods_electrical_propulsion_pod; assumption_resolution: component: energy_harvesting_mpp t_conversion_mppt_power_tracker; assumption_resolution: component: energy_harvesting_solar_skin_array_solar_cell_array; assumption_resolution:	pending; pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.
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Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
					component: structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure; assumption_resolution: sanity:haps_endurance_target; plus 23 more			

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 3; numeric_fact_count : 3; requirement_trace_covered: 3; requirement_trace_partial: 0; requirement_trace_uncovered: 0	wingspan_m: 35 m; endurance_days: 30 days; altitude_km: 20 km; wingspan_m: covered; endurance_days: covered; altitude_km: covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected haps with high confidence from brief keywords.	selected_class: haps; confidence: high; keyword_scores: haps:6	haps: 6	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 11-module haps architecture from the scratch universal architecture grammar.	module_count: 11; architecture_source : scratch_universal_architecture	Ultra-Light Wing Structure; Solar Energy Harvesting; Night-Cycle Energy Storage; Flight Power Distribution; Propulsion And Control Surfaces; Autonomy Control Communication; plus 5 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 33; average_submodules_per_module: 3	structure_containment.wing_primary_structure; structure_containment.tail_control_structure; structure_containment.payload_bay_structure; energy_harvesting.solar_skin_array; energy_harvesting.mppt_conversion; energy_harvesting.solar_health_monitoring; plus 27 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 8; present_links: 8; missing_links: 0; graph_nodes: 44; graph_edges: 88; shared_interface_edges: 47; missing_required_graph_edges: 0	energy_harvesting -> power_distribution via solar_dc_bus: present; energy_storage_source -> power_distribution via battery_dc_bus: present; power_distribution -> actuation_kinematics via propulsion_power_b	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
				us: present; control_compute_communication -> actuation_kinematics via flight_control_bus: present; sensing_instrumentation -> control_compute_communication via sensor_bus: present; payload_communication -> control_compute_communication via payload_data_link: present; plus 2 more	
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 132; critical_candidate_lines: 12; admitted_price_lines: 0	high_aspect_ratio_wing_structure: High-aspect-ratio wing structure; carbon_spar_cap_set: carbon spar cap set; foam_rib_set: foam rib set; wing_joiner_sleeve: wing joiner sleeve; tail_boom_assembly: tail boom assembly; elevator_surface: elevator surface; plus 24 more	Showing first 30 of 132 component candidates in the stage trace.
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_count: 0; sanity_pass: 5; sanity_warn: 0; sanity_fail: 0	pass Stratospheric Altitude Band: 20 km; pass Station-Keeping Endurance: 30 days; pass Wing Span Envelope: 35 m; pass Solar/Energy/Flight Closure: present; pass Payload/Telemetry/Safety Closure: present	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign-off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 132; admitted_lines: 0; unpriced_lines: 132; unpriced_critical_lines: 12; rejected_source_records: 0	none	BoM cost remains intentionally blocked until 12 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

11 modules, 33 sub-modules, 132 allocated component words.

From	To	Interface	Status	Engineering Reason
energy_harvesting	power_distribution	solar_dc_bus	OK	Solar array output must feed power management over a declared DC bus.
energy_storage_source	power_distribution	battery_dc_bus	OK	Night-cycle battery energy must connect into load distribution.
power_distribution	actuation_kinematics	propulsion_power_bus	OK	Propulsion loads must receive protected high-current power.
control_compute_communication	actuation_kinematics	flight_control_bus	OK	Flight controller must command propulsion and trim actuators.
sensing_instrumentation	control_compute_communication	sensor_bus	OK	Navigation, airdata and structural state must feed autonomy decisions.
payload_communication	control_compute_communication	payload_data_link	OK	Payload state must be visible to vehicle autonomy and telemetry.
environmental_interface	energy_storage_source	thermal_path	OK	Battery survival and night-cycle capacity depend on a declared thermal path.
safety_protection	control_compute_communication	telemetry_link	OK	Abort, descent and recovery state must remain visible to ground operators.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
11	33	47	8	0

Interface Contract Matrix

8/8 required contracts present. 17 shared interface families, 1 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
solar_dc_bus	Solar Energy Harvesting	Solar skin array; MPPT conversion; Solar health monitoring	Flight Power Distribution	Solar and battery power management	present	Solar array output must feed power management over a declared DC bus.	Both endpoints expose the required interface through at least one submodule carrier.
battery_dc_buses	Night-Cycle Energy Storage	Stratospheric battery pack	Flight Power Distribution	Solar and battery power management	present	Night-cycle battery energy must connect into load distribution.	Both endpoints expose the required interface through at least one submodule carrier.
propulsion_power_bus	Flight Power Distribution	Solar and battery power management	Propulsion And Control Surfaces	Electric propulsion pods	present	Propulsion loads must receive protected high-current power.	Both endpoints expose the required interface through at least one submodule carrier.
flight_control_bus	Autonomy Control Communication	Flight control computer; Energy-aware autonomy	Propulsion And Control Surfaces	Electric propulsion pods; Control surface actuation; Station-keeping trim	present	Flight controller must command propulsion and trim actuators.	Both endpoints expose the required interface through at least one submodule carrier.
sensor_bus	Navigation And Airdata Sensing	Navigation suite; Airdata and weather sensors; Structural health sensing	Autonomy Control Communication	Flight control computer	present	Navigation, airdata and structural state must feed autonomy decisions.	Both endpoints expose the required interface through at least one submodule carrier.
payload_data_link	Communications Payload	Payload radio; Antenna aperture	Autonomy Control Communication	Ground command link; Energy-aware autonomy	present	Payload state must be visible to vehicle autonomy and telemetry.	Both endpoints expose the required interface through at least one submodule carrier.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
thermal_path	Stratospheric Environmental Interface	Thermal protection; Low-pressure electronics management	Night-Cycle Energy Storage	Stratospheric battery pack; Battery thermal zone	present	Battery survival and night-cycle capacity depend on a declared thermal path.	Both endpoints expose the required interface through at least one submodule carrier.
telemetry_link	Flight Safety And Recovery	Abort and recovery; Airspace safety	Autonomy Control Communication	Ground command link	present	Abort, descent and recovery state must remain visible to ground operators.	Both endpoints expose the required interface through at least one submodule carrier.

Local-only declarations to review later: payload_mount (Ultra-Light Wing Structure).

Interface Verification Gate

interface_review_ready

8 required interface row(s): 0 accepted, 8 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
solar_dc_bus	Solar Energy Harvesting	Flight Power Distribution	both_endpoints_carried	interface_review:energy_harvesting:power_distribution:solar_dc_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
battery_dc_buses	Night-Cycle Energy Storage	Flight Power Distribution	both_endpoints_carried	interface_review:energy_storage_source:power_distribution:battery_dc_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
propulsion_power_bus	Flight Power Distribution	Propulsion And Control Surfaces	both_endpoints_carried	interface_review:power_distribution:actuation_kinematics:propulsion_power_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
flight_control_bus	Autonomy Control Communication	Propulsion And Control Surfaces	both_endpoints_carried	interface_review:control_compute_communication:actuation_kinematics:flight_control_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
sensor_bus	Navigation And Airdata Sensing	Autonomy Control Communication	both_endpoints_carried	interface_review:sensing_instrumentation:control_compute_communication:sensor_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
payload_data_link	Communications Payload	Autonomy Control Communication	both_endpoints_carried	interface_review:payload_communication:control_compute_communication:payload_data_link	pending	review_ready	Collect named interface-review evidence through verification intake.
thermal_path	Stratospheric Environmental Interface	Night-Cycle Energy Storage	both_endpoints_carried	interface_review:environmental_interface:energy_storage_source:thermal_path	pending	review_ready	Collect named interface-review evidence through verification intake.
telemetry_link	Flight Safety And Recovery	Autonomy Control Communication	both_endpoints_carried	interface_review:safety_protection:control_compute_communication:telemetry_link	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_ready

33/33 submodules ready; 0 review, 0 blocked. Component words 132; critical parts 12/12; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Ultra-Light Wing Structure	Wing primary structure	ready	4	aero_load_path; solar_mount; propulsion_mounts	high_aspect_ratio_wing_structure	none	Submodule has component candidates and no allocation blocker.
Ultra-Light Wing Structure	Tail and control structure	ready	4	aero_load_path; flight_control_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Ultra-Light Wing Structure	Payload bay structure	ready	4	payload_mount; service_access; aero_load_path	none	none	Submodule has component candidates and no allocation blocker.
Solar Energy Harvesting	Solar skin array	ready	4	solar_dc_bus; solar_mount; thermal_path	solar_cell_array	none	Submodule has component candidates and no allocation blocker.
Solar Energy Harvesting	MPPT conversion	ready	4	solar_dc_bus; sensor_bus; service_access	mppt_power_tracker	none	Submodule has component candidates and no allocation blocker.
Solar Energy Harvesting	Solar health monitoring	ready	4	solar_dc_bus; sensor_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Night-Cycle Energy Storage	Stratospheric battery pack	ready	4	battery_dc_buses; thermal_path; battery_monitor_bus	stratospheric_battery_pack	none	Submodule has component candidates and no allocation blocker.
Night-Cycle Energy Storage	Battery thermal zone	ready	4	thermal_path; battery_monitor_bus; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Night-Cycle Energy Storage	Energy reserve monitoring	ready	4	battery_monitor_bus; sensor_bus; telemetry_link	none	none	Submodule has component candidates and no allocation blocker.
Flight Power Distribution	Solar and battery power management	ready	4	solar_dc_bus; battery_dc_buses; propulsion_power_bus; payload_power_bus	none	none	Submodule has component candidates and no allocation blocker.
Flight Power Distribution	Low-voltage distribution	ready	4	low_voltage_rail; payload_power_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Flight Power Distribution	Load-shed protection	ready	4	hardwired_trip; payload_power_bus; battery_monitor_bus	none	none	Submodule has component candidates and no allocation blocker.
Propulsion And Control Surfaces	Electric propulsion pods	ready	4	propulsion_power_bus; flight_control_bus; propulsion_mounts	electric_propulsion_pod	none	Submodule has component candidates and no allocation blocker.
Propulsion And Control Surfaces	Control surface actuation	ready	4	flight_control_bus; aero_load_path; low_voltage_rail	none	none	Submodule has component candidates and no allocation blocker.
Propulsion And Control Surfaces	Station-keeping trim	ready	4	flight_control_bus; sensor_bus; aero_load_path	none	none	Submodule has component candidates and no allocation blocker.
Autonomy Control Communication	Flight control computer	ready	4	flight_control_bus; sensor_bus; battery_monitor_bus; hardwired_trip	flight_control_computer	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Autonomy Control Communication	Ground command link	ready	4	telemetry_link ; payload_data_link; service_access	ground_control_link	none	Submodule has component candidates and no allocation blocker.
Autonomy Control Communication	Energy-aware autonomy	ready	4	battery_monitor_bus; payload_data_link; flight_control_bus	none	none	Submodule has component candidates and no allocation blocker.
Navigation And Airdata Sensing	Navigation suite	ready	4	sensor_bus; flight_control_bus; telemetry_link	gnss_ins_navigation_unit	none	Submodule has component candidates and no allocation blocker.
Navigation And Airdata Sensing	Airdata and weather sensors	ready	4	sensor_bus; flight_control_bus	none	none	Submodule has component candidates and no allocation blocker.
Navigation And Airdata Sensing	Structural health sensing	ready	4	sensor_bus; aero_load_path; telemetry_link	none	none	Submodule has component candidates and no allocation blocker.
Communications Payload	Payload radio	ready	4	payload_data_link; payload_power_bus; thermal_path	stratospheric_communications_payload	none	Submodule has component candidates and no allocation blocker.
Communications Payload	Antenna aperture	ready	4	rf_aperture; payload_data_link; aero_load_path	payload_antenna_array	none	Submodule has component candidates and no allocation blocker.
Communications Payload	Payload integration	ready	4	payload_power_bus; thermal_path; service_access	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Stratospheric Environmental Interface	Thermal protection	ready	4	thermal_path; service_access	thermal_insulation_blanket	none	Submodule has component candidates and no allocation blocker.
Stratospheric Environmental Interface	UV and surface protection	ready	4	solar_mount; aero_load_path; service_access	none	none	Submodule has component candidates and no allocation blocker.
Stratospheric Environmental Interface	Low-pressure electronics management	ready	4	thermal_path; payload_power_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Flight Safety And Recovery	Abort and recovery	ready	4	hardwired_trip; telemetry_link; flight_control_bus	recovery_parachute_system	none	Submodule has component candidates and no allocation blocker.
Flight Safety And Recovery	Airspace safety	ready	4	flight_control_bus; telemetry_link; hardwired_trip	none	none	Submodule has component candidates and no allocation blocker.
Flight Safety And Recovery	Ground handling safety	ready	4	service_access; aero_load_path; battery_monitor_bus	none	none	Submodule has component candidates and no allocation blocker.
Ground Handling Serviceability	Wing assembly service	ready	4	service_access; aero_load_path; solar_mount	none	none	Submodule has component candidates and no allocation blocker.
Ground Handling Serviceability	Launch and recovery equipment	ready	4	service_access; aero_load_path; telemetry_link	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Ground Handling Serviceability	Payload checkout	ready	4	service_access; payload_data_link; telemetry_link	none	none	Submodule has component candidates and no allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/33 submodules pass; 33 review, 0 blocked. Component words 132; linked requirements 13; carrier contracts 28/28; local-only interfaces 71. Review questions accepted 0/94; verification accepted 0/83; critical unpriced lines 12.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Ultra-Light Wing Structure	Wing primary structure	review	4	3	1	0/0	0/2	0/3	1	0.6667	Review 3 local-only interface declaration(s): aero_load_path, solar_mount, propulsion_mounts. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Ultra-Light Wing Structure	Tail and control structure	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): aero_load_path, flight_control_buss, service_access. 2 engineering

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ring review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Ultra-Light Wing Structure	Payload bay structure	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): payload_mount, service_access, aero_load_path. 2 engineer ring review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Solar Energy Harvesting	Solar skin array	review	4	3	2	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): solar_mount, thermal_path. 3 engineer ring review question(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											n(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Solar Energy Harvesting	MPPT conversion	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration(s): sensor_ bus, service_ access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											cost/manufactur er/MPN evidence.
Solar Energy Harvesting	Solar health monitoring	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): sensor_bus, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Night-Cycle Energy Storage	Stratospheric battery pack	review	4	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): battery_monitor_bus. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											component line(s) still need source-backed cost/manufacturer/MPN evidence.
Night-Cycle Energy Storage	Battery thermal zone	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): battery_monitor_bus, sensor_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Night-Cycle Energy Storage	Energy reserve monitoring	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): battery_monitor_bus, sensor_bus, telemetry_link. 2 engineering review question(s) await acceptance.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											nce. 1 verification activity row(s) await accepted evidence.
Flight Power Distribution	Solar and battery power management	review	4	4	1	3/3	0/5	0/5	0	0.6667	Review 1 local-only interface declaration(s): payload_power_bus. 5 engineering review question(s) await acceptance. 5 verification activity row(s) await accepted evidence.
Flight Power Distribution	Low-voltage distribution	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): low_voltage_rail, payload_power_bus, service_access. 2 engineering review question(s) await acceptance. 1 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ion activity row(s) await accepted evidence.
Flight Power Distribution	Load-shed protection	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): hardwired_trip, payload_power_bus, battery_monitor_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Propulsion And Control Surfaces	Electric propulsion pods	review	4	3	2	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): propulsion_mounts. 4 engineering review question(s) await acceptance. 4 verification activity row(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Propulsion And Control Surfaces	Control surface actuation	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): aero_load_path, low_voltage_rail. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Propulsion And Control Surfaces	Station-keeping trim	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): sensor_bus, aero_load_path. 3 engineering

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Autonomy Control Communication	Flight control computer	review	4	4	2	2/2	0/4	0/5	1	0.6667	Review 2 local-only interfaces declaration(s): battery_monitor_bus, hardwired_trip. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Autonomy Control	Ground command link	review	4	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Communication											e declarat ion(s): service_ access. 4 enginee ring review questio n(s) await accepta nce. 3 verificat ion activity row(s) await accepte d evidenc e. 1 critical compon ent line(s) still need source- backed cost/ma nufactur er/MPN evidenc e.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Autonomy Control Communication	Energy-aware autonomy	review	4	3	0	2/2	0/4	0/3	0	0.6667	Review 1 local-only interface declaration(s): battery_monitor_bus. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence.
Navigation And Airdata Sensing	Navigation suite	review	4	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): flight_control_bus, telemetry_link. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											source-backed cost/manufacturer/MPN evidence.
Navigation And Airdata Sensing	Airdata and weather sensors	review	4	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): flight_control_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Navigation And Airdata Sensing	Structural health sensing	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): aero_load_path, telemetry_link. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
Communications Payload	Payload radio	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration(s): payload_power_bus, thermal_path. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await acceptance. 1 evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Communications Payload	Antenna aperture	review	4	3	0	1/1	0/3	0/3	1	0.6667	Review 2 local-only interface declaration(s): rf_aperture, aero_load_path. 3 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Communications Payload	Payload integration	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): payload_power_bus, thermal_path, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Stratospheric Environmental Interface	Thermal protection	review	4	2	1	1/1	0/3	0/4	1	0.6667	Review 1 local-only interface declaration(s): service_access. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await acceptance. 1 evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.

Stratospheric Environmental Interface	UV and surface protection	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): solar_mount, aero_load_path, service_access. 2 engineering review question(s) await acceptance.
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Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											nce. 1 verification activity row(s) await accepted evidence.
Stratospheric Environmental Interface	Low-pressure electronics management	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): payload_power_bus, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Flight Safety And Recovery	Abort and recovery	review	4	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): hardwired_trip, flight_control_buss. 3 engineering review question(s) await acceptance. 3 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Flight Safety And Recovery	Airspace safety	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): flight_control_b us, hardwired_trip. 3 engineer review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Flight Safety And Recovery	Ground handling safety	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, aero_load_path ,

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											battery_monitor_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Ground Handling Serviceability	Wing assembly service	review	4	3	1	0/0	0/2	0/2	0	0.6667	Review 3 local-only interface declaration(s): service_access, aero_load_path, solar_mount. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Ground Handling Serviceability	Launch and recovery equipment	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, aero_load_path

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											, telemetry_link. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Ground Handling Serviceability	Payload checkout	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, payload_data_link, telemetry_link. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
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Module Engineering Gate

module_engineering_review_required

0/11 modules pass; 11 review, 0 blocked. Submodules/components: 33/132. Interface carriers 16/16; review questions accepted 0/116; critical unpriced lines 12 across 9 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Ultra-Light Wing Structure	review	3	12	1	0/0	0/8	1	0.86	1 critical BoM line(s) need source-backed cost evidence. 8 engineering review question(s) await acceptance. 3 assumption row(s) need review or source evidence.
Solar Energy Harvesting	review	3	12	2	1/1	0/12	2	0.86	2 critical BoM line(s) need source-backed cost evidence. 12 engineering review question(s) await acceptance. 8 assumption row(s) need review or source evidence.
Night-Cycle Energy Storage	review	3	12	1	2/2	0/10	1	0.86	1 critical BoM line(s) need source-backed cost evidence. 10 engineering review question(s) await acceptance. 5 assumption row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Flight Power Distributio n	review	3	12	1	3/3	0/10	0	0.86	10 engineerin g review question(s) await acceptanc e. 5 assumptio n row(s) need review or source evidence.
Propulsion And Control Surfaces	review	3	12	2	2/2	0/12	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 12 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Autonomy Control Communic ation	review	3	12	2	4/4	0/17	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 17 engineerin g review question(s) await acceptanc e. 10 assumptio n row(s) need review or source evidence.

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Navigation And Airdata Sensing	review	3	12	1	1/1	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 7 assumptio n row(s) need review or source evidence.
Communic ations Payload	review	3	12	0	1/1	0/7	2	0.71	2 critical BoM line(s) need source- backed cost evidence. 7 engineerin g review question(s) await acceptanc e. 4 assumptio n row(s) need review or source evidence.
Stratosph eric Environme ntal Interface	review	3	12	1	1/1	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 5 assumptio n row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Flight Safety And Recovery	review	3	12	1	1/1	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Ground Handling Serviceabi lity	review	3	12	1	0/0	0/7	0	0.86	7 engineerin g review question(s) await acceptanc e. 3 assumptio n row(s) need review or source evidence.

Module Review Roll-Up

2 ready, 9 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Ultra-Light Wing Structure	sourcing	3	12	0/0	wingspan_m	high_aspect_ratio_wing_structure	1 critical line(s) need source-backed evidence before BoM pass.
Solar Energy Harvesting	sourcing	3	12	1/1	wingspan_m; endurance_days	solar_cell_array; mppt_power_tracker	2 critical line(s) need source-backed evidence before BoM pass.
Night-Cycle Energy Storage	sourcing	3	12	2/2	endurance_days	stratospheric_battery_pack	1 critical line(s) need source-backed evidence before BoM pass.
Flight Power Distribution	ready	3	12	3/3	endurance_days	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Propulsion And Control Surfaces	sourcing	3	12	2/2	wingspan_m; endurance_days	electric_propulsion_pod	1 critical line(s) need source-backed evidence before BoM pass.
Autonomy Control Communication	sourcing	3	12	4/4	endurance_days; altitude_km	flight_control_computer; ground_control_link	2 critical line(s) need source-backed evidence before BoM pass.
Navigation And Airdata Sensing	sourcing	3	12	1/1	altitude_km	gnss_ins_navigation_unit	1 critical line(s) need source-backed evidence before BoM pass.
Communications Payload	sourcing	3	12	1/1	none	stratospheric_communications_payload; payload_antenna_array	2 critical line(s) need source-backed evidence before BoM pass.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Stratospheric Environmental Interface	sourcing	3	12	1/1	altitude_km	thermal_insulation_blanket	1 critical line(s) need source-backed evidence before BoM pass.
Flight Safety And Recovery	sourcing	3	12	1/1	altitude_km	recovery_parachute_system	1 critical line(s) need source-backed evidence before BoM pass.
Ground Handling Serviceability	ready	3	12	0/0	wingspan_m	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Engineering Review Pack

86 reviewer questions: 0 accepted, 53 ready, 33 needing review, 0 blocked. Coverage: 11 modules, 33 submodules, 8 interfaces, 3 calculations, 31 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumptio n_resoluti on	major	needs review	CAA-UAS-OPS	Can the CAA-UAS-OPS assumption be accepted for architecture review, or must it be replaced with evidence?	Airspace concept of operations , detect-and-avoid assumptions, command link and recovery evidence.	none	none	none	none
assumptio n_resoluti on	major	needs review	DO-160-ENV	Can the DO-160-ENV assumption be accepted for architecture review, or must it be replaced with evidence?	Temperatu re, vibration, electrical power and EMC environmental evidence tailored to stratosphe ric operation.	none	none	none	none
assumptio n_resoluti on	major	needs review	SOLAR-ENERGY-BALANCE-REVIEW	Can the SOLAR-ENERGY-BALANCE-REVIEW assumption be accepted for architecture review, or must it be replaced with evidence?	Solar collection, battery night-cycle, propulsion load and station-keeping wind model evidence.	none	none	none	none
assumptio n_resoluti on	major	needs review	Electric propulsion pod	Can the Electric propulsion pod assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	actuation_kinematics	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	Flight control computer	Can the Flight control computer assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	Ground control link	Can the Ground control link assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	MPPT power tracker	Can the MPPT power tracker assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_harvesting	none	none	none
assumption_resolution	major	needs review	Solar cell array	Can the Solar cell array assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_harvesting	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	Stratospheric battery pack	Can the Stratospheric battery pack assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_storage_source	none	none	none
assumption_resolution	major	needs review	Thermal insulation blanket	Can the Thermal insulation blanket assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none
assumption_resolution	major	needs review	Payload antenna array	Can the Payload antenna array assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	payload_communication	none	none	none
assumption_resolution	major	needs review	Stratospheric communications payload	Can the Stratospheric communications payload assumption be accepted for architecture review, or must it be replaced	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	payload_communication	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				with evidence?					
assumption_resolution	major	needs review	Recovery parachute system	Can the Recovery parachute system assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	safety_protection	none	none	none
assumption_resolution	major	needs review	GNSS INS navigation unit	Can the GNSS INS navigation unit assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	High-aspect-ratio wing structure	Can the High-aspect-ratio wing structure assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	structure_containment	none	none	none
assumption_resolution	major	needs review	control_compute_communication -> actuation_kinematics	Can the control_compute_communication -> actuation_kinematics assumption be accepted for architecture review,	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed	none	flight_control_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				or must it be replaced with evidence?	design or sourcing.				
assumption_resolution	major	needs review	energy_harvesting -> power_distribution	Can the energy_harvesting -> power_distribution assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	solar_dc_bus	none	none
assumption_resolution	major	needs review	energy_storage_source -> power_distribution	Can the energy_storage_source -> power_distribution assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	battery_dc_bus	none	none
assumption_resolution	major	needs review	environmental_interface -> energy_storage_source	Can the environmental_interface -> energy_storage_source assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	thermal_path	none	none
assumption_resolution	major	needs review	payload_communication -> control_co	Can the payload_communication ->	Review the interface carrier,	none	payload_data_link	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
			mpute_co mmunicati on	control_co mpute_co mmunicati on assumptio n be accepted for architecture review, or must it be replaced with evidence?	direction, capacity, safety margin and failure mode before detailed design or sourcing.				
assumptio n_resoluti on	major	needs review	power_dis tribution - > actuation_ kinematics	Can the power_dis tribution - > actuation_ kinematics assumptio n be accepted for architectu re review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	propulsion _power_b us	none	none
assumptio n_resoluti on	major	needs review	safety_pro tection -> control_co mpute_co mmunicati on	Can the safety_pro tection -> control_co mpute_co mmunicati on assumptio n be accepted for architectu re review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	telemetry_ link	none	none
assumptio n_resoluti on	major	needs review	sensing_in strumenta tion -> control_co mpute_co mmunicati on	Can the sensing_in strumenta tion -> control_co mpute_co mmunicati on assumptio n be accepted for	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed	none	sensor_bu s	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				architectu re review, or must it be replaced with evidence?	design or sourcing.				
assumptio n_resoluti on	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumptio n be accepted for architectu re review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumptio n_resoluti on	major	needs review	Station-keeping endurance	Can the Station-keeping endurance assumptio n be accepted for architectu re review, or must it be replaced with evidence?	Review the formula, input values and operating envelope before treating this metric as a performan ce claim.	none	none	none	none
assumptio n_resoluti on	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumptio n be accepted for architectu re review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumptio n_resoluti on	major	needs review	Stratosph eric Altitude Band	Can the Stratosph eric Altitude Band assumptio n be accepted for architectu	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale	environme ntal_interf ace; sensing_in strumenta tion; control_co mpute_co mmunicati on;	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				re review, or must it be replaced with evidence?	for this envelope.	safety_protection			
assumption_resolution	major	needs review	Station-Keeping Endurance	Can the Station-Keeping Endurance assumption be accepted for architectural review, or must it be replaced with evidence?	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.	energy_harvesting; energy_storage_source; power_distribution; actuation_kinematics; control_compute_communication	none	none	none
assumption_resolution	major	needs review	Solar/Energy/Flight Closure	Can the Solar/Energy/Flight Closure assumption be accepted for architectural review, or must it be replaced with evidence?	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.	structure_containment; energy_harvesting; actuation_kinematics; maintenance_serviceability; energy_storage_source; power_distribution; control_compute_communication; environmental_interface; sensing_instrumentation; safety_protection	none	none	none
assumption_resolution	major	needs review	Payload/Telemetry/Safety Closure	Can the Payload/Telemetry/Safety Closure assumption be accepted for architectural review, or must it be	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.	environmental_interface; sensing_instrumentation; control_compute_communication; safety_protection	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				replaced with evidence?					
assumption_resolution	major	needs review	Wing Span Envelope	Can the Wing Span Envelope assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.	structure_containment; energy_harvesting; actuation_kinematics; maintenance_serviceability	none	none	none
calculation_envelope	major	needs review	HAPS endurance hours	Does the HAPS endurance hours calculation support the architecture envelope, or does the concept need revision?	Confirm solar resource, battery reserve, wind field, degradation, payload power and recovery plan.	energy_harvesting; energy_storage_source; power_distribution; actuation_kinematics; control_compute_communication	none	haps_endurance_hours	none
calculation_envelope	major	needs review	HAPS wingspan-to-altitude marker	Does the HAPS wingspan-to-altitude marker calculation support the architecture envelope, or does the concept need revision?	Confirm wing area, mass budget, airfoil, aeroelastic margin, launch loads and transport constraints.	structure_containment; energy_harvesting; actuation_kinematics; maintenance_serviceability; environmental_interface; sensing_instrumentation; control_compute_communication; safety_protection	none	haps_wingspan_to_altitude_ratio	none
calculation_envelope	major	ready	HAPS altitude conversion	Does the HAPS altitude conversion calculation support	Confirm operating altitude, airspace permissions,	environmental_interface; sensing_instrumentation;	none	haps_altitude_ft	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				the architecture envelope, or does the concept need revision?	atmospheric density, thermal envelope and wind model.	control_compute_communication; safety_protection			
interface_contract	major	ready	Autonomy Control Communication -> Propulsion And Control Surfaces via flight_control_bus	Is the flight_control_bus interface between Autonomy Control Communication and Propulsion And Control Surfaces correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	control_compute_communication; actuation_kinematics	flight_control_bus	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Solar Energy Harvesting -> Flight Power Distribution via solar_dc_bus	Is the solar_dc_bus interface between Solar Energy Harvesting and Flight Power Distribution correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_harvesting; power_distribution	solar_dc_bus	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Night-Cycle Energy Storage -> Flight Power Distribution via battery_dc_bus	Is the battery_dc_bus interface between Night-Cycle Energy Storage and Flight Power Distribution correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_storage_source; power_distribution	battery_dc_bus	none	Both endpoints expose the required interface through at least one submodule carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
interface_contract	major	ready	Stratospheric Environmental Interface -> Night-Cycle Energy Storage via thermal_path	Is the thermal_path interface between Stratospheric Environmental Interface and Night-Cycle Energy Storage correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	environmental_interface; energy_storage_source	thermal_path	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Communications Payload -> Autonomy Control Communication via payload_data_link	Is the payload_data_link interface between Communications Payload and Autonomy Control Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	payload_communication; control_computation	payload_data_link	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Flight Power Distribution -> Propulsion And Control Surfaces via propulsion_power_buss	Is the propulsion_power_buss interface between Flight Power Distribution and Propulsion And Control Surfaces correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	power_distribution; actuation_kinematics	propulsion_power_buss	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Flight Safety And Recovery ->	Is the telemetry_link interface between	Interface-review evidence naming both	safety_protection; control_computation	telemetry_link	none	Both endpoints expose the required

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
			Autonomy Control Communication via telemetry_link	Flight Safety And Recovery and Autonomy Control Communication correctly owned and physically carried?	endpoint owners, carrier submodules, capacity limits and any deferred checks.	mmunication			interface through at least one submodule carrier.
interface_contract	major	ready	Navigation And Airdata Sensing - > Autonomy Control Communication via sensor_buses	Is the sensor_buses interface between Navigation And Airdata Sensing and Autonomy Control Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	sensing_instrumentation; control_compute_communication	sensor_buses	none	Both endpoints expose the required interface through at least one submodule carrier.
module_allocation	major	ready	Propulsion And Control Surfaces	Does Propulsion And Control Surfaces have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	actuation_kinematics	propulsion_power_buses; flight_control_bus; propulsion_mounts; aero_load_path; low_voltage_rail; sensor_buses	haps_wingspan_envelope; haps_energy_flight_closure; haps_endurance_target	none
module_allocation	major	ready	Autonomy Control Communication	Does Autonomy Control Communication have the right responsibility boundary, submodule split and interface ownership	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	control_compute_communication	flight_control_bus; sensor_buses; telemetry_link; battery_monitor_bus; ; payload_data_link; hardwired_trip;	haps_endurance_target; haps_energy_flight_closure; haps_altitude_band; haps_payload_commands_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				for this concept?			service_access		
module_allocation	major	ready	Solar Energy Harvesting	Does Solar Energy Harvesting have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	energy_harvesting	solar_dc_bus; solar_mount; sensor_buses; thermal_path; service_access	haps_wing_span_envelope; haps_energy_flight_closure; haps_endurance_target	none
module_allocation	major	ready	Night-Cycle Energy Storage	Does Night-Cycle Energy Storage have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	energy_storage_source	battery_dc_bus; thermal_path; battery_monitor_bus; service_access; sensor_buses; telemetry_link	haps_endurance_target; haps_energy_flight_closure	none
module_allocation	major	ready	Stratospheric Environmental Interface	Does Stratospheric Environmental Interface have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	environmental_interface	thermal_path; solar_mount; service_access; rf_aperture; aero_load_path; payload_power_bus	haps_altitude_band; haps_energy_flight_closure; haps_payload_comms_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Ground Handling Serviceability	Does Ground Handling Serviceability have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	maintenance_service_ability	service_access; aero_load_path; telemetry_link; solar_mount; payload_data_link	haps_wing_span_envelope; haps_energy_flight_closure	none
module_all ocation	major	ready	Communications Payload	Does Communications Payload have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	payload_communication	payload_data_link; payload_power_bus; rf_aperture; thermal_path; service_access; aero_load_path	none	none
module_all ocation	major	ready	Flight Power Distribution	Does Flight Power Distribution have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	power_distribution	solar_dc_bus; battery_dc_bus; propulsion_power_bus; payload_power_bus; low_voltage_rail; hardwired_trip; service_access; battery_monitor_bus	haps_endurance_target; haps_energy_flight_closure	none
module_all ocation	major	ready	Flight Safety And Recovery	Does Flight Safety And Recovery have the right responsibility boundary, submodule split and interface ownership	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	safety_protection	hardwired_trip; telemetry_link; flight_control_bus; service_access; aero_load_path; battery_monitor_bus	haps_altitude_band; haps_energy_flight_closure; haps_payload_comms_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for this concept?					
module_all ocation	major	ready	Navigation And Airdata Sensing	Does Navigation And Airdata Sensing have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	sensing_in strumentation	sensor_bu s; flight_cont rol_bus; aero_load _path; telemetry_ link	haps_altit ude_band; haps_ener gy_flight_ closure; haps_payl oad_com ms_safety	none
module_all ocation	major	ready	Ultra-Light Wing Structure	Does Ultra-Light Wing Structure have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	structure_ containme nt	aero_load _path; solar_mou nt; propulsion _mounts; service_ac cess; payload_ mount; flight_cont rol_bus	haps_wing span_enve lope; haps_ener gy_flight_ closure	none
submodul e_allocatio n	major	ready	Propulsion And Control Surfaces / Control surface actuation	Do the componen t candidate s and interfaces assigned to Control surface actuation make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	actuation_ kinematics	flight_cont rol_bus; aero_load _path; low_voltag e_rail	haps_wing span_enve lope; haps_ener gy_flight_ closure; haps_end urance_tar get	none
submodul e_allocatio n	major	ready	Propulsion And Control Surfaces / Electric propulsion pods	Do the componen t candidate s and interfaces assigned to Electric propulsion pods make engineering sense	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or	actuation_ kinematics	propulsion _power_b us; flight_cont rol_bus; propulsion _mounts	haps_wing span_enve lope; haps_ener gy_flight_ closure; haps_end urance_tar get	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for its stated purpose?	capacity checks.				
submodul e_allocation	major	ready	Propulsion And Control Surfaces / Station-keeping trim	Do the componen t candidate s and interfaces assigned to Station-keeping trim make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	actuation_kinematics	flight_cont rol_bus; sensor_bu s; aero_load _path	haps_wing span_enve lope; haps_ener gy_flight_closure; haps_end urance_tar get	none
submodul e_allocation	major	ready	Autonomy Control Communication / Energy-aware autonomy	Do the componen t candidate s and interfaces assigned to Energy-aware autonomy make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	battery_m onitor_bus ; payload_d ata_link; flight_cont rol_bus	haps_end urance_tar get; haps_ener gy_flight_closure; haps_altit ude_band; haps_payl oad_com ms_safety	none
submodul e_allocation	major	ready	Autonomy Control Communication / Flight control computer	Do the componen t candidate s and interfaces assigned to Flight control computer make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	flight_cont rol_bus; sensor_bu s; battery_m onitor_bus ; hardwired _trip	haps_end urance_tar get; haps_ener gy_flight_closure; haps_altit ude_band; haps_payl oad_com ms_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Autonomy Control Communication / Ground command link	Do the component candidates and interfaces assigned to Ground command link make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	telemetry_link; payload_data_link; service_access	haps_endurance_target; haps_energy_flight_closure; haps_altitude_band; haps_payload_comms_safety	none
submodule_allocation	major	ready	Solar Energy Harvesting / MPPT conversion	Do the component candidates and interfaces assigned to MPPT conversion make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_harvesting	solar_dc_bus; sensor_buses; service_access	haps_wing_span_envelope; haps_energy_flight_closure; haps_endurance_target	none
submodule_allocation	major	ready	Solar Energy Harvesting / Solar health monitoring	Do the component candidates and interfaces assigned to Solar health monitoring make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_harvesting	solar_dc_bus; sensor_buses; service_access	haps_wing_span_envelope; haps_energy_flight_closure; haps_endurance_target	none
submodule_allocation	major	ready	Solar Energy Harvesting / Solar skin array	Do the component candidates and interfaces assigned to Solar skin array make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_harvesting	solar_dc_bus; solar_mount; thermal_path	haps_wing_span_envelope; haps_energy_flight_closure; haps_endurance_target	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Night-Cycle Energy Storage / Battery thermal zone	Do the component candidates and interfaces assigned to Battery thermal zone make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	thermal_path; battery_monitor_bus; sensor_buses	haps_endurance_target; haps_energy_flight_closure	none
submodule_allocation	major	ready	Night-Cycle Energy Storage / Energy reserve monitoring	Do the component candidates and interfaces assigned to Energy reserve monitoring make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	battery_monitor_bus; sensor_buses; telemetry_link	haps_endurance_target; haps_energy_flight_closure	none
submodule_allocation	major	ready	Night-Cycle Energy Storage / Stratospheric battery pack	Do the component candidates and interfaces assigned to Stratospheric battery pack make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	battery_dc_bus; thermal_path; battery_monitor_bus	haps_endurance_target; haps_energy_flight_closure	none
submodule_allocation	major	ready	Stratospheric Environmental Interface / Low-pressure electronics management	Do the component candidates and interfaces assigned to Low-pressure electronics management make	Reviewer note confirming the submodule role, component allocation and any missing carriers or	environmental_interface	thermal_path; payload_power_bus; service_access	haps_altitude_band; haps_energy_flight_closure; haps_payload_comms_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Stratospheric Environmental Interface / Thermal protection	engineering sense for its stated purpose?	capacity checks.				
				Do the component candidate s and interfaces assigned to Thermal protection make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	thermal_path; service_access	haps_altitude_band; haps_energy_flight_closure; haps_payload_commands_safety	none
submodule_allocation	major	ready	Stratospheric Environmental Interface / UV and surface protection	Do the component candidate s and interfaces assigned to UV and surface protection make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	solar_mount; aero_load_path; service_access	haps_altitude_band; haps_energy_flight_closure; haps_payload_commands_safety	none
submodule_allocation	major	ready	Ground Handling Serviceability / Launch and recovery equipment	Do the component candidate s and interfaces assigned to Launch and recovery equipment make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_serviceability	service_access; aero_load_path; telemetry_link	haps_wing_span_envelope; haps_energy_flight_closure	none
submodule_allocation	major	ready	Ground Handling Serviceability / Payload checkout	Do the component candidate s and interfaces assigned to Payload	Reviewer note confirming the submodule role, component	maintenance_serviceability	service_access; payload_data_link; telemetry_link	haps_wing_span_envelope; haps_energy_flight_closure	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				checkout make engineering sense for its stated purpose?	allocation and any missing carriers or capacity checks.				

Showing first 70 of 86 engineering review questions; export CSV for the full pack.

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
Stratospheric Altitude Band	pass	20 km	15 to 25 km for common HAPS station-keeping concepts; outside this band needs atmospheric, power and airspace evidence	20 km sits in the stratospheric operating band before detailed wind, airspace and thermal environment evidence.
Station-Keeping Endurance	pass	30 days	<=60 days plausible for early solar HAPS concept, 60-180 days needs strong degradation and energy evidence, >180 days is high risk	30 days requires solar day/night balance, degradation allowance, battery reserve and stratospheric wind validation.
Wing Span Envelope	pass	35 m	10 to 80 m for lightweight demonstrator-to-service HAPS concepts before aeroelastic proof	35 m is plausible for a solar HAPS wing, but still needs aeroelastic, transport and launch-handling evidence.
Solar/Energy/Flight Closure	pass	present	solar DC, battery DC, propulsion power, flight control and battery monitor paths present	Solar generation, night-cycle battery storage, propulsion loads and flight control are connected before BoM sourcing.
Payload/Telemetry/Safety Closure	pass	present	payload data, payload power, telemetry, hardwired trip, thermal path and recovery parachute present	Payload relay, ground command, thermal protection and recovery paths are named before sourcing or flight-readiness claims.

Engineering Calculation Ledger

3 calculations: 1 within deterministic envelope, 2 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
HAPS endurance hours	needs review	endurance_days * 24	720 h	Arithmetic only; station keeping needs wind, day/night energy and degradation proof.	720 h endurance target before stratospheric energy-balance proof.	Confirm solar resource, battery reserve, wind field, degradation, payload power and recovery plan.
HAPS altitude conversion	within envelope	altitude_km * 3,280.84	65616.8 ft	49,000 to 82,000 ft is a broad HAPS operating band before airspace proof.	65616.8 ft converted from target altitude.	Confirm operating altitude, airspace permissions, atmospheric density, thermal envelope and wind model.
HAPS wingspan-to-altitude marker	needs review	wingspan_m / altitude_km	1.75 m/km	Marker only; aerodynamic feasibility needs wing area, mass, Reynolds number and aeroelastic proof.	1.75 m/km is a trace marker, not an aerodynamic proof.	Confirm wing area, mass budget, airfoil, aeroelastic margin, launch loads and transport constraints.

Engineering Assumption Ledger

34 assumptions: 3 brief-supported, 8 model-present, 9 needing engineering review, 14 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 14.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Wingspan	brief_requirement	brief supported	no	no	Wingspan = 35 m is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Endurance	brief_requirement	brief supported	no	no	Endurance = 30 days is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Operating altitude	brief_requirement	brief supported	no	no	Operating altitude = 20 km is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Station-keeping endurance	derived_metric	review required	no	no	Station-keeping endurance uses endurance_days with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Stratospheric Altitude Band	sanity_envelope	review required	no	no	Stratospheric Altitude Band currently evaluates as pass against 15 to 25 km for common HAPS station-keeping concepts; outside this band needs atmospheric, power and	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					airspace evidence.	
Station-Keeping Endurance	sanity_envelope	review required	no	no	Station-Keeping Endurance currently evaluates as pass against <=60 days plausible for early solar HAPS concept, 60-180 days needs strong degradation and energy evidence, >180 days is high risk.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Wing Span Envelope	sanity_envelope	review required	no	no	Wing Span Envelope currently evaluates as pass against 10 to 80 m for lightweight demonstrator-to-service HAPS concepts before aeroelastic proof.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Solar/Energy/Flight Closure	sanity_envelope	review required	no	no	Solar/Energy/Flight Closure currently evaluates as pass against solar DC, battery DC, propulsion power, flight control and battery monitor paths present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Payload/Telemetry/Safety Closure	sanity_envelope	review required	no	no	Payload/Telemetry/Safety Closure currently evaluates as pass against payload data, payload power, telemetry, hardwired trip, thermal path and recovery parachute present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
energy_harvesting -> power_distribution	interface_closure	model present	no	no	solar_dc_bus is named on both endpoint modules in the	Review the interface carrier, direction, capacity, safety margin and

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					generated architecture.	failure mode before detailed design or sourcing.
energy_storage_source -> power_distribution	interface_closure	model present	no	no	battery_dc_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
power_distribution -> actuation_kinematics	interface_closure	model present	no	no	propulsion_power_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_compute_communication -> actuation_kinematics	interface_closure	model present	no	no	flight_control_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
sensing_instrumentation -> control_compute_communication	interface_closure	model present	no	no	sensor_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
payload_communication -> control_compute_communication	interface_closure	model present	no	no	payload_data_link is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
environmental_interface -> energy_storage_source	interface_closure	model present	no	no	thermal_path is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
safety_protection -> control_computer_communication	interface_closure	model present	no	no	telemetry_link is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
High-aspect-ratio wing structure	critical_component	source required	no	yes	High-aspect-ratio wing structure is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Solar cell array	critical_component	source required	no	yes	Solar cell array is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
MPPT power tracker	critical_component	source required	no	yes	MPPT power tracker is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Stratospheric battery pack	critical_component	source required	no	yes	Stratospheric battery pack is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Electric propulsion pod	critical_component	source required	no	yes	Electric propulsion pod is a critical architecture-derived component candidate with quantity 4 each.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Flight control computer	critical_component	source required	no	yes	Flight control computer is a critical	Admit supplier, manufacturer, MPN where

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					architecture-derived component candidate with quantity 1 candidate.	applicable, unit cost, lead time and evidence reference through sourcing intake.
Ground control link	critical_component	source required	no	yes	Ground control link is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
GNSS INS navigation unit	critical_component	source required	no	yes	GNSS INS navigation unit is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Stratospheric communications payload	critical_component	source required	no	yes	Stratospheric communications payload is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Payload antenna array	critical_component	source required	no	yes	Payload antenna array is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Thermal insulation blanket	critical_component	source required	no	yes	Thermal insulation blanket is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Recovery parachute system	critical_component	source required	no	yes	Recovery parachute system is a critical architecture-	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					derived component candidate with quantity 1 candidate.	and evidence reference through sourcing intake.
CAA-UAS-OPS	compliance_evidence	review required	no	no	High-altitude unmanned aircraft operational safety case is relevant to this product class in UK / aviation authority review.	Airspace concept of operations, detect-and-avoid assumptions, command link and recovery evidence.
DO-160-ENV	compliance_evidence	review required	no	no	Airborne equipment environmental conditions is relevant to this product class in Aviation engineering reference.	Temperature, vibration, electrical power and EMC environmental evidence tailored to stratospheric operation.
SOLAR-ENERGY-BALANCE-REVIEW	compliance_evidence	review required	no	no	Solar day/night energy balance review is relevant to this product class in Project engineering gate.	Solar collection, battery night-cycle, propulsion load and station-keeping wind model evidence.

Engineering Verification Plan

44 activities: 29 ready for review, 6 open, 9 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Ultra-Light Wing Structure architecture allocation	ready	design_review	Ultra-Light Wing Structure	wingspan_m	aero_load_path; solar_mount; propulsion_mounts; service_access; payload_mount; flight_control_bus	high_aspect_ratio_wing_structure; carbon_spacer_cap_set; foam_rib_set; wing_joiner_sleeve; tail_boom_assembly; elevator_surface; rudder_surface; control_hinge_set; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Ultra-Light Wing Structure requirement calculations	ready	calculation	Ultra-Light Wing Structure	wingspan_m	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Ultra-Light Wing Structure critical components	blocked	source_evidence	Ultra-Light Wing Structure	wingspan_m	none	high_aspect_ratio_wing_structure	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	high_aspect_ratio_wing_structure

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Solar Energy Harvesting architecture allocation	ready	design_review	Solar Energy Harvesting	wingspan_m; endurance_days	solar_dc_buses; solar_mount; sensor_bus; thermal_path; service_access	solar_cell_array; cell_interconnect_ribbon; transparent_encapsulation_film; solar_string_bypass_diode; mppt_power_tracker; solar_input_current_sensor; solar_voltage_tap_loom; mppt_heat_spreader; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Solar Energy Harvesting requirement calculations	ready	calculation	Solar Energy Harvesting	wingspan_m; endurance_days	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Solar Energy Harvesting critical components	blocked	source_evidence	Solar Energy Harvesting	wingspan_m; endurance_days	none	solar_cell_array; mppt_power_tracker	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	solar_cell_array mppt_power_tracker
Review Night-Cycle Energy Storage architecture allocation	ready	design_review	Night-Cycle Energy Storage	endurance_days	battery_dc_bus; thermal_path; battery_monitor_bus; service_access; sensor_bus; telemetry_link	stratospheric_battery_pack; battery_management_board; lightweight_battery_enclosure; cell_heater_film; battery_insulation_sleeve; phase_change_thermal_buffer; battery_temperature_sensors_harness; battery_vent_path; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Night-Cycle Energy Storage requirement calculations	ready	calculation	Night-Cycle Energy Storage	endurance_days	none	none	Each linked requirement has a calculation, sanity check or clearly	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Night-Cycle Energy Storage critical components	blocked	source_evidence	Night-Cycle Energy Storage	endurance_days	none	stratospheric_battery_pack	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	stratospheric_battery_pack
Review Flight Power Distribution architecture allocation	ready	design_review	Flight Power Distribution	endurance_days	solar_dc_buses; battery_dc_bus; propulsion_power_bus; payload_power_bus; low_voltage_rail; hardwired_trip; service_access; battery_monitor_bus	solar_battery_charge_controller; main_dc_buses_board; payload_power_switch; propulsion_bus_fuseset; avionics_dc_converter; servo_power_rail; communications_dc_converter; low_voltage_harness_lo	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						om; plus 4 more	No blocker or major architecture issue is attached to the module.	
Verify Flight Power Distribution requirement calculations	ready	calculation	Flight Power Distribution	endurance_days	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Review Propulsion And Control Surfaces architecture allocation	ready	design_review	Propulsion And Control Surfaces	wingspan_m; endurance_days	propulsion_power_bus; flight_control_bus; propulsion_mounts; aero_load_path; low_voltage_rail; sensor_bus	electric_propulsion_pod; high_altitude_propeller_set; motor_controller_module; propulsion_pod_fairing; low_temperature_servo_actuator; control_linkage_rod_set; surface_position_sensor; servo_heater_trace; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Propulsion And Control Surfaces requirement calculations	ready	calculation	Propulsion And Control Surfaces	wingspan_m; endurance_days	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Propulsion And Control Surfaces critical components	blocked	source_evidence	Propulsion And Control Surfaces	wingspan_m; endurance_days	none	electric_propulsion_pod	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	electric_propulsion_pod
Review Autonomy Control Communication architecture allocation	ready	design_review	Autonomy Control Communication	endurance_days; altitude_km	flight_control_bus; sensor_bus; telemetry_link; battery_monitor_bus; payload_data_link; hardwired_trip; service_access	flight_control_computer; autopilot_io_board; flight_software_image; watchdog_supervisor; ground_control_link; telemetry_transceiver; command_authentication_module; mission_log_recorder; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							architecture issue is attached to the module.	
Verify Autonomy Control Communication requirement calculations	ready	calculation	Autonomy Control Communication	endurance_days; altitude_km	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Autonomy Control Communication critical components	blocked	source_evidence	Autonomy Control Communication	endurance_days; altitude_km	none	flight_control_computer; ground_control_link	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	flight_control_computer ground_control_link
Review Navigation And Airdata Sensing architecture allocation	ready	design_review	Navigation And Airdata Sensing	altitude_km	sensor_bus; flight_control_bus; aero_load_path; telemetry_link	gnss_ins_navigation_unit; barometric_altitude_sensor; sun_sensor_pair;	Every submodule has a stated purpose and at least one component candidate.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						precision_tim mebase_mod ule; pitot_static _probe; outside_air_ temperature _probe; wind_estim ation_filter; humidity_fr ost_sensor; plus 4 more	Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Verify Navigation And Airdata Sensing requirement calculations	ready	calculation	Navigation And Airdata Sensing	altitude_km	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source- backed evidence for Navigation And Airdata Sensing critical components	blocked	source_evid ence	Navigation And Airdata Sensing	altitude_km	none	gnss_ins_na vigation_uni t	Supplier name, manufactur er or equivalent, MPN where applicable, unit cost and source URL/referen ce are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component	gnss_ins_na vigation_uni t

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							identity match the architecture candidate line.	
Review Communications Payload architecture allocation	ready	design_review	Communications Payload	none	payload_data_link; payload_power_bus; rf_aperture; thermal_path; service_access; aero_load_path	stratospheric_communications_payload; payload_modem_board; payload_rf_power_amplifier; payload_processor_module; payload_antenna_array; antenna_ground_plane_film; antenna_feed_harness; beam_pointing_calibration_target; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Collect source-backed evidence for Communications Payload critical components	blocked	source_evidence	Communications Payload	none	none	stratospheric_communications_payload; payload_antenna_array	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	stratospheric_communications_payload payload_antenna_array

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Stratospheric Environmental Interface architecture allocation	ready	design_review	Stratospheric Environmental Interface	altitude_km	thermal_patching; solar_mounting; service_access; rf_aperture; aero_load_path; payload_power_bus	thermal_insulation_blanket; multi_layer_insulation_patch; lightweight_thermal_strip; thermal_witness_label; uv_protective_film; leading_edge_erosion_tape; surface_contamination_wipe_kit; coating_inspection_marker; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Stratospheric Environmental Interface requirement calculations	ready	calculation	Stratospheric Environmental Interface	altitude_km	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Stratospheric Environmental Interface critical components	blocked	source_evidence	Stratospheric Environmental Interface	altitude_km	none	thermal_insulation_blanket	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	thermal_insulation_blanket

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Flight Safety And Recovery architecture allocation	ready	design_review	Flight Safety And Recovery	altitude_km	hardwired_trip; telemetry_link; flight_control_bus; service_access; aero_load_path; battery_monitor_bus	recovery_parachute_system; parachute_deployment_controller; recovery_beacon; safe_descent_mode_logic; geofence_database; lost_link_failsafe; flight_termination_relay; airspace_transponder_module; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Flight Safety And Recovery requirement calculations	ready	calculation	Flight Safety And Recovery	altitude_km	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Flight Safety And Recovery critical components	blocked	source_evidence	Flight Safety And Recovery	altitude_km	none	recovery_parachute_system	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	recovery_parachute_system
Review Ground Handling Serviceability architecture allocation	ready	design_review	Ground Handling Serviceability	wingspan_m	service_access; aero_load_path; telemetry_link; solar_mount; payload_data_link	wing_assembly_jig; spar_pin_tool; solar_skin_protection_mat; assembly_witness_mark_set; launch_dolly_interface; wingtip_handling_fixture; recovery_cradle; ground_wind_limit_placard; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Ground Handling Serviceability requirement calculations	ready	calculation	Ground Handling Serviceability	wingspan_m	none	none	Each linked requirement has a calculation, sanity check or clearly	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Verify solar_dc_buss interface contract	ready	interface_review	Solar Energy Harvesting -> Flight Power Distribution	none	solar_dc_buss	solar_cell_array; cell_interconnect_ribbon; transparent_encapsulation_film; solar_string_bypass_diode; mppt_power_tracker; solar_input_current_sensor; solar_voltage_tap_loom; mppt_heat_spreader; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify battery_dc_bus interface contract	ready	interface_review	Night-Cycle Energy Storage -> Flight Power Distribution	none	battery_dc_bus	stratospheric_battery_pack; battery_management_board; lightweight_battery_enclosure; cell_heater_film; solar_battery_charge_controller; main_dc_buss_board; payload_power_switch; propulsion_bus_fuse_set	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify propulsion_power_bus	ready	interface_review	Flight Power Distribution -> Propulsion	none	propulsion_power_bus	solar_battery_charge_controller; main_dc_bu	Both endpoint modules declare the	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
interface contract			And Control Surfaces			s_board; payload_power_switch; propulsion_bus_fuse_set; electric_propulsion_pod; ; high_altitude_propeller_set; motor_controller_module; propulsion_pod_fairing	interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	
Verify flight_control_bus interface contract	ready	interface_review	Autonomy Control Communication -> Propulsion And Control Surfaces	none	flight_control_bus	flight_control_computer; autopilot_io_board; flight_software_image; watchdog_supervisor; solar_forecast_model; wind_field_route_planner; payload_duty_cycle_controller; return_corridor_manager; plus 12 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify sensor_bus interface contract	ready	interface_review	Navigation And Airdata Sensing -> Autonomy Control Communication	none	sensor_bus	gnss_ins_navigation_unit; barometric_altitude_sensor; sun_sensor_pair; precision_timebase_module; pitot_static_probe; outside_air_temperature_probe; wind_estimation_filter; humidity_frost_sensor; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Verify payload_data_link interface contract	ready	interface_review	Communications Payload -> Autonomy Control Communication	none	payload_data_link	stratospheric_communications_payload; payload_modem_board; payload_rf_power_amplifier; payload_processor_module; payload_antenna_array; antenna_ground_plane_film; antenna_feed_harness; beam_pointing_calibration_target; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify thermal_path interface contract	ready	interface_review	Stratospheric Environmental Interface -> Night-Cycle Energy Storage	none	thermal_path	thermal_insulation_blanket; multi_layer_insulation_patch; lightweight_thermal_strip; thermal_witness_label; low_pressure_vent_path; conformal_coating_set; high_altitude_creepage_spacer; desiccant_cartridge; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify telemetry_link interface contract	ready	interface_review	Flight Safety And Recovery -> Autonomy Control Communication	none	telemetry_link	recovery_parachute_system; parachute_deployment_controller; recovery_beacon; safe_descent_mode_logic; geofence_database; lost_link_failsafe; flight_termination_relay	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						; airspace_transponder_module; plus 4 more	validated by the next detailed design pass.	
Plan evidence for CAA-UAS-OPS	open	compliance_review	System Compliance	none	none	none	Airspace concept of operations, detect-and-avoid assumptions, command link and recovery evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for DO-160-ENV	open	compliance_review	System Compliance	none	none	none	Temperature, vibration, electrical power and EMC environmental evidence tailored to stratospheric operation.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for SOLAR-ENERGY-BALANCE-REVIEW	open	compliance_review	System Compliance	none	none	none	Solar collection, battery night-cycle, propulsion load and station-keeping wind model evidence.	Compliance evidence not yet collected in scratch prototype.
Review risk control for Energy-negative overnight operation	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.
Review risk control for	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence	Risk evidence

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Wing structural failure in gust or launch handling							and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	not yet reviewed in scratch prototype.
Review risk control for Lost command link or uncontrolled descent	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 35 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Ultra-Light Wing Structure architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Ultra-Light Wing Structure requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Ultra-Light Wing Structure critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Solar Energy Harvesting architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Solar Energy Harvesting requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Solar Energy Harvesting critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Night-Cycle Energy Storage architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Night-Cycle Energy Storage requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Night-Cycle Energy Storage critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Flight Power Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Flight Power Distribution requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Review Propulsion And Control Surfaces architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify Propulsion And Control Surfaces requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Propulsion And Control Surfaces critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Autonomy Control Communication architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Autonomy Control Communication requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Autonomy Control Communication critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Navigation And Airdata Sensing architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Navigation And Airdata Sensing requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Navigation And Airdata Sensing critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Communications Payload architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Communications Payload critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Stratospheric Environmental Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify Stratospheric Environmental Interface requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Stratospheric Environmental Interface critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Flight Safety And Recovery architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Flight Safety And Recovery requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Flight Safety And Recovery critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Ground Handling Serviceability architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Ground Handling Serviceability requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Verify solar_dc_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify battery_dc_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify propulsion_power_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify flight_control_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify sensor_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify payload_data_link interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify thermal_path interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify telemetry_link interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Plan evidence for CAA-UAS-OPS	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for DO-160-ENV	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for SOLAR-ENERGY-BALANCE-REVIEW	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Energy-negative overnight operation	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Wing structural failure in gust or launch handling	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Lost command link or uncontrolled descent	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Ultra-Light Wing Structure

Carries a roughly 35 m high-aspect-ratio wing, solar skin, propulsion pods and payload loads.

aero_load_path	solar_mount	propulsion_mounts	service_access	payload_mount
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Wing primary structure

Provides the long-span lift structure and distributed payload mounting.

Components: High-aspect-ratio wing structure, carbon spar cap set, foam rib set, wing joiner sleeve

Tail and control structure

Provides static stability and trim surfaces.

Components: tail boom assembly, elevator surface, rudder surface, control hinge set

Payload bay structure

Supports payload, avionics and access panels while preserving aerodynamic form.

Components: payload bay frame, avionics tray, access panel latch set, mass balance bracket

Solar Energy Harvesting

Converts daytime stratospheric sunlight into regulated electrical energy.

solar_dc_bus	solar_mount	sensor_bus	thermal_path
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Solar skin array

Distributes photovoltaic cells over wing surfaces.

Components: Solar cell array, cell interconnect ribbon, transparent encapsulation film, solar string bypass diode

MPPT conversion

Tracks photovoltaic maximum power and feeds the flight DC bus.

Components: MPPT power tracker, solar input current sensor, solar voltage tap loom, MPPT heat spreader

Solar health monitoring

Detects string degradation and weathering across long endurance missions.

Components: solar string monitor board, UV exposure witness coupon, wing surface temperature sensor, solar inspection connector

Night-Cycle Energy Storage

Stores enough solar energy for night operation during a 30 day station-keeping target.

battery_dc_bus	thermal_path	battery_monitor_bus	service_access
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Stratospheric battery pack

Stores night-cycle and transient propulsion energy.

Components: Stratospheric battery pack, battery management board, lightweight battery enclosure, cell heater film

Battery thermal zone

Protects battery capacity in low-pressure, cold stratospheric conditions.

Components: battery insulation sleeve, phase-change thermal buffer, battery temperature sensor harness, battery vent path

Energy reserve monitoring

Tracks day/night balance and mission abort reserve.

Components: coulomb counter module, reserve energy estimator, battery isolation contactor, night-cycle test connector

Flight Power Distribution

Balances solar input, battery storage and protected load rails for propulsion, avionics and payload.

solar_dc_bus	battery_dc_bus	propulsion_power_bus	payload_power_bus	low_voltage_rail	hardwired_trip
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Solar and battery power management

Combines harvested and stored energy into mission load rails.

Components: solar battery charge controller, main DC bus board, payload power switch, propulsion bus fuse set

Low-voltage distribution

Feeds avionics, sensors, servos and communications loads.

Components: avionics DC converter, servo power rail, communications DC converter, low-voltage harness loom

Load-shed protection

Protects mission reserve during energy-negative conditions.

Components: load shed relay set, brownout supervisor, energy priority controller, fault latch indicator

Propulsion And Control Surfaces

Generates efficient station-keeping thrust and aerodynamic trim in thin stratospheric air.

propulsion_power_bus	flight_control_bus	propulsion_mounts	aero_load_path
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Electric propulsion pods

Provides distributed propulsive thrust along the wing.

Components: Electric propulsion pod, high-altitude propeller set, motor controller module, propulsion pod fairing

Control surface actuation

Moves trim surfaces with low power and reliable feedback.

Components: low-temperature servo actuator, control linkage rod set, surface position sensor, servo heater trace

Station-keeping trim

Supports efficient loiter in variable stratospheric winds.

Components: trim schedule table, gust load estimator, airspeed setpoint manager, propulsion calibration record

Autonomy Control Communication

Runs autonomous station keeping, energy management, payload coordination and ground command links.

flight_control_bus	sensor_bus	telemetry_link	battery_monitor_bus	payload_data_link	hardwired_trip
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Flight control computer

Executes flight stabilisation, energy-aware routing and station keeping.

Components: Flight control computer, autopilot IO board, flight software image, watchdog supervisor

Ground command link

Maintains command, telemetry and mission supervision.

Components: Ground control link, telemetry transceiver, command authentication module, mission log recorder

Energy-aware autonomy

Coordinates route, payload duty cycle and energy reserve.

Components: solar forecast model, wind field route planner, payload duty-cycle controller, return corridor manager

Navigation And Airdata Sensing

Measures navigation, airdata, structural loads and energy state for closed-loop autonomy.

- sensor_bus
- flight_control_bus
- aero_load_path
- telemetry_link

Navigation suite

Provides position, attitude and timing references.

Components: GNSS INS navigation unit, barometric altitude sensor, sun sensor pair, precision timebase module

Airdata and weather sensors

Measures local atmosphere and flight condition.

Components: pitot static probe, outside air temperature probe, wind estimation filter, humidity frost sensor

Structural health sensing

Tracks load and deformation over long-span flight.

Components: wing strain gauge strip, spar temperature sensor, vibration accelerometer, structural health recorder

Communications Payload

Provides relay or observation service payload with controlled power, RF and data interfaces.

- payload_data_link
- payload_power_bus
- rf_aperture
- thermal_path
- service_access

Payload radio

Carries communications relay payload electronics.

Components: Stratospheric communications payload, payload modem board, payload RF power amplifier, payload processor module

Antenna aperture

Provides RF aperture and beam orientation support.

Components: Payload antenna array, antenna ground plane film, antenna feed harness, beam pointing calibration target

Payload integration

Controls payload mounting, thermal coupling and service access.

Components: payload isolation mount, payload thermal strap, payload access hatch, payload configuration tag

Stratospheric Environmental Interface

Protects systems from low pressure, cold soak, UV exposure and thermal cycling near 20 km altitude.

- thermal_path
- solar_mount
- service_access
- rf_aperture
- aero_load_path

Thermal protection

Controls cold-soak and daytime heating across batteries, avionics and payload.

Components: Thermal insulation blanket, multi-layer insulation patch, lightweight thermal strap, thermal witness label

UV and surface protection

Protects solar skin and airframe surface from UV, ozone and abrasion.

Components: UV protective film, leading edge erosion tape, surface contamination wipe kit, coating inspection marker

Low-pressure electronics management

Accounts for low pressure cooling and electrical spacing.

Components: low-pressure vent path, conformal coating set, high-altitude creepage spacer, desiccant cartridge

Flight Safety And Recovery

Protects airspace, ground risk and vehicle hardware during energy faults, lost-link and descent.

hardwired_trip	telemetry_link	flight_control_bus	service_access	aero_load_path
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Abort and recovery

Provides controlled descent and recovery location after mission abort.

Components: Recovery parachute system, parachute deployment controller, recovery beacon, safe descent mode logic

Airspace safety

Maintains geofence, lost-link and flight termination policies.

Components: geofence database, lost-link failsafe, flight termination relay, airspace transponder module

Ground handling safety

Controls launch, recovery and battery handling hazards.

Components: launch arming interlock, propulsion safety pin set, battery handling placard, recovery inspection checklist

Ground Handling Serviceability

Supports wing assembly, launch preparation, payload checkout and recovery inspection.

service_access	aero_load_path	telemetry_link	solar_mount
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Wing assembly service

Makes distributed wing sections inspectable and assembled repeatably.

Components: wing assembly jig, spar pin torque tool, solar skin protection mat, assembly witness mark set

Launch and recovery equipment

Supports low-speed launch and controlled recovery handling.

Components: launch dolly interface, wingtip handling fixture, recovery cradle, ground wind limit placard

Payload checkout

Verifies communications payload and telemetry before launch.

Components: payload checkout cable, RF load test adapter, telemetry ground terminal, preflight log card

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: £0

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
High-aspect-ratio wing structure	1	candidate	unpriced	unpriced				
carbon spar cap set	1	candidate	unpriced	unpriced				
foam rib set	1	candidate	unpriced	unpriced				
wing joiner sleeve	1	candidate	unpriced	unpriced				
tail boom assembly	1	candidate	unpriced	unpriced				
elevator surface	1	candidate	unpriced	unpriced				
rudder surface	1	candidate	unpriced	unpriced				
control hinge set	1	candidate	unpriced	unpriced				
payload bay frame	1	candidate	unpriced	unpriced				
avionics tray	1	candidate	unpriced	unpriced				
access panel latch set	1	candidate	unpriced	unpriced				
mass balance bracket	1	candidate	unpriced	unpriced				
Solar cell array	1	candidate	unpriced	unpriced				
cell interconnect ribbon	1	candidate	unpriced	unpriced				
transparent encapsulation film	1	candidate	unpriced	unpriced				
solar string bypass diode	1	candidate	unpriced	unpriced				
MPPT power tracker	1	candidate	unpriced	unpriced				
solar input current sensor	1	candidate	unpriced	unpriced				
solar voltage tap	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
loom								
MPPT heat spreader	1	candidate	unpriced	unpriced				
solar string monitor board	1	candidate	unpriced	unpriced				
UV exposure witness coupon	1	candidate	unpriced	unpriced				
wing surface temperature sensor	1	candidate	unpriced	unpriced				
solar inspection connector	1	candidate	unpriced	unpriced				
Stratospheric battery pack	1	candidate	unpriced	unpriced				
battery management board	1	candidate	unpriced	unpriced				
lightweight battery enclosure	1	candidate	unpriced	unpriced				
cell heater film	1	candidate	unpriced	unpriced				
battery insulation sleeve	1	candidate	unpriced	unpriced				
phase-change thermal buffer	1	candidate	unpriced	unpriced				
battery temperature sensor harness	1	candidate	unpriced	unpriced				
battery vent path	1	candidate	unpriced	unpriced				
coulomb counter module	1	candidate	unpriced	unpriced				
reserve energy estimator	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
battery isolation contactor	1	candidate	unpriced	unpriced				
night-cycle test connector	1	candidate	unpriced	unpriced				
solar battery charge controller	1	candidate	unpriced	unpriced				
main DC bus board	1	candidate	unpriced	unpriced				
payload power switch	1	candidate	unpriced	unpriced				
propulsion bus fuse set	1	candidate	unpriced	unpriced				
avionics DC converter	1	candidate	unpriced	unpriced				
servo power rail	1	candidate	unpriced	unpriced				
communications DC converter	1	candidate	unpriced	unpriced				
low-voltage harness loom	1	candidate	unpriced	unpriced				
load shed relay set	1	candidate	unpriced	unpriced				
brownout supervisor	1	candidate	unpriced	unpriced				
energy priority controller	1	candidate	unpriced	unpriced				
fault latch indicator	1	candidate	unpriced	unpriced				
Electric propulsion pod	4	each	unpriced	unpriced				
high-altitude propeller set	1	candidate	unpriced	unpriced				
motor controller module	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
propulsion pod fairing	1	candidate	unpriced	unpriced				
low-temperature servo actuator	1	candidate	unpriced	unpriced				
control linkage rod set	1	candidate	unpriced	unpriced				
surface position sensor	1	candidate	unpriced	unpriced				
servo heater trace	1	candidate	unpriced	unpriced				
trim schedule table	1	candidate	unpriced	unpriced				
gust load estimator	1	candidate	unpriced	unpriced				
airspeed setpoint manager	1	candidate	unpriced	unpriced				
propulsion calibration record	1	candidate	unpriced	unpriced				
Flight control computer	1	candidate	unpriced	unpriced				
autopilot IO board	1	candidate	unpriced	unpriced				
flight software image	1	candidate	unpriced	unpriced				
watchdog supervisor	1	candidate	unpriced	unpriced				
Ground control link	1	candidate	unpriced	unpriced				
telemetry transceiver	1	candidate	unpriced	unpriced				
command authentication module	1	candidate	unpriced	unpriced				
mission log recorder	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
solar forecast model	1	candidate	unpriced	unpriced				
wind field route planner	1	candidate	unpriced	unpriced				
payload duty-cycle controller	1	candidate	unpriced	unpriced				
return corridor manager	1	candidate	unpriced	unpriced				
GNSS INS navigation unit	1	candidate	unpriced	unpriced				
barometric altitude sensor	1	candidate	unpriced	unpriced				
sun sensor pair	1	candidate	unpriced	unpriced				
precision timebase module	1	candidate	unpriced	unpriced				
pitot static probe	1	candidate	unpriced	unpriced				
outside air temperature probe	1	candidate	unpriced	unpriced				
wind estimation filter	1	candidate	unpriced	unpriced				
humidity frost sensor	1	candidate	unpriced	unpriced				
wing strain gauge strip	1	candidate	unpriced	unpriced				
spar temperature sensor	1	candidate	unpriced	unpriced				
vibration acceleromet er	1	candidate	unpriced	unpriced				
structural health recorder	1	candidate	unpriced	unpriced				
Stratospheri c communicat	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
ions payload								
payload modem board	1	candidate	unpriced	unpriced				
payload RF power amplifier	1	candidate	unpriced	unpriced				
payload processor module	1	candidate	unpriced	unpriced				
Payload antenna array	1	candidate	unpriced	unpriced				
antenna ground plane film	1	candidate	unpriced	unpriced				
antenna feed harness	1	candidate	unpriced	unpriced				
beam pointing calibration target	1	candidate	unpriced	unpriced				
payload isolation mount	1	candidate	unpriced	unpriced				
payload thermal strap	1	candidate	unpriced	unpriced				
payload access hatch	1	candidate	unpriced	unpriced				
payload configuratio n tag	1	candidate	unpriced	unpriced				
Thermal insulation blanket	1	candidate	unpriced	unpriced				
multi-layer insulation patch	1	candidate	unpriced	unpriced				
lightweight thermal strap	1	candidate	unpriced	unpriced				
thermal witness label	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
UV protective film	1	candidate	unpriced	unpriced				
leading edge erosion tape	1	candidate	unpriced	unpriced				
surface contamination wipe kit	1	candidate	unpriced	unpriced				
coating inspection marker	1	candidate	unpriced	unpriced				
low-pressure vent path	1	candidate	unpriced	unpriced				
conformal coating set	1	candidate	unpriced	unpriced				
high-altitude creepage spacer	1	candidate	unpriced	unpriced				
desiccant cartridge	1	candidate	unpriced	unpriced				
Recovery parachute system	1	candidate	unpriced	unpriced				
parachute deployment controller	1	candidate	unpriced	unpriced				
recovery beacon	1	candidate	unpriced	unpriced				
safe descent mode logic	1	candidate	unpriced	unpriced				
geofence database	1	candidate	unpriced	unpriced				
lost-link failsafe	1	candidate	unpriced	unpriced				
flight termination relay	1	candidate	unpriced	unpriced				
airspace transponder module	1	candidate	unpriced	unpriced				
launch arming interlock	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
propulsion safety pin set	1	candidate	unpriced	unpriced				
battery handling placard	1	candidate	unpriced	unpriced				
recovery inspection checklist	1	candidate	unpriced	unpriced				
wing assembly jig	1	candidate	unpriced	unpriced				
spar pin torque tool	1	candidate	unpriced	unpriced				
solar skin protection mat	1	candidate	unpriced	unpriced				
assembly witness mark set	1	candidate	unpriced	unpriced				
launch dolly interface	1	candidate	unpriced	unpriced				
wingtip handling fixture	1	candidate	unpriced	unpriced				
recovery cradle	1	candidate	unpriced	unpriced				
ground wind limit placard	1	candidate	unpriced	unpriced				
payload checkout cable	1	candidate	unpriced	unpriced				
RF load test adapter	1	candidate	unpriced	unpriced				
telemetry ground terminal	1	candidate	unpriced	unpriced				
preflight log card	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	132	0	132	12

Critical Sourcing Worklist

Component	Qty	Reason
High-aspect-ratio wing structure	1 candidate	Required class-critical line must be sourced before BoM can pass.
Solar cell array	1 candidate	Required class-critical line must be sourced before BoM can pass.
MPPT power tracker	1 candidate	Required class-critical line must be sourced before BoM can pass.
Stratospheric battery pack	1 candidate	Required class-critical line must be sourced before BoM can pass.
Electric propulsion pod	4 each	Required class-critical line must be sourced before BoM can pass.
Flight control computer	1 candidate	Required class-critical line must be sourced before BoM can pass.
Ground control link	1 candidate	Required class-critical line must be sourced before BoM can pass.
GNSS INS navigation unit	1 candidate	Required class-critical line must be sourced before BoM can pass.
Stratospheric communications payload	1 candidate	Required class-critical line must be sourced before BoM can pass.
Payload antenna array	1 candidate	Required class-critical line must be sourced before BoM can pass.
Thermal insulation blanket	1 candidate	Required class-critical line must be sourced before BoM can pass.
Recovery parachute system	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/132 priced lines admitted. Critical coverage 0/12. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWorldId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
High-aspect-ratio wing structure	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
carbon spar cap set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
foam rib set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
wing joiner sleeve	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
tail boom assembly	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
elevator surface	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
rudder surface	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
control hinge set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
payload bay frame	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
avionics tray	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
access panel latch set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
mass balance bracket	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
Solar cell array	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
cell interconnect ribbon	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
transparent encapsulation film	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
solar string bypass diode	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
MPPT power tracker	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
solar input current sensor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									when detailed procurement planning starts.
solar voltage tap loom	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
MPPT heat spreader	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
solar string monitor board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
UV exposure witness coupon	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
wing surface temperature sensor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufact urer	MPN	Evidence Ref	Next Action
solar inspection connector	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Stratospheric battery pack	critical unpriced	unique	critical	unpriced					Collect admissible supplier/ manufacturer/ MPN/unit -cost evidence before BoM can pass.
battery management board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
lightweight battery enclosure	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cell heater film	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery insulation sleeve	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
phase-change thermal buffer	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery temperature sensor harness	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery vent path	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
coulomb counter module	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
reserve energy estimator	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
battery isolation contactor	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
night-cycle test connector	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
solar battery charge controller	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
main DC bus board	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
payload power switch	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
propulsion bus fuse set	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Showing first 40 of 132 ledger rows; export CSV for the full line-by-line ledger.

0 source-backed claims, 48 missing-source claims, 0 provenance violations. Critical missing-source claims: 48. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
High-aspect-ratio wing structure	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
High-aspect-ratio wing structure	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
High-aspect-ratio wing structure	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
High-aspect-ratio wing structure	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Solar cell array	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Solar cell array	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Solar cell array	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Solar cell array	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MPPT power tracker	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MPPT power tracker	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MPPT power tracker	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
MPPT power tracker	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Stratospheric battery pack	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Stratospheric battery pack	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Stratospheric battery pack	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Stratospheric battery pack	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Electric propulsion pod	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Electric propulsion pod	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Electric propulsion pod	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
					evidence before BoM can pass.
Electric propulsion pod	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight control computer	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight control computer	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight control computer	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Flight control computer	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Ground control link	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Ground control link	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Ground control link	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Ground control link	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
GNSS INS navigation unit	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
GNSS INS navigation unit	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
GNSS INS navigation unit	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
GNSS INS navigation unit	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Stratospheric communications payload	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Stratospheric communications payload	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Stratospheric communications payload	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Stratospheric communications payload	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Payload antenna array	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Payload antenna array	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Payload antenna array	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Payload antenna array	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thermal insulation blanket	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thermal insulation blanket	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thermal insulation blanket	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thermal insulation blanket	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Recovery parachute system	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Recovery parachute system	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Recovery parachute system	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Recovery parachute system	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Source Reference Quality Gate

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/132 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 12 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
High-aspect-ratio wing structure	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
carbon spar cap set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
foam rib set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
wing joiner sleeve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
tail boom assembly	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
elevator surface	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
rudder surface	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
control hinge set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
payload bay frame	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
avionics tray	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
access panel latch set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
mass balance bracket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Solar cell array	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
cell interconnect ribbon	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
transparent encapsulation film	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
solar string bypass diode	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
MPPT power tracker	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									evidence for this critical line before BoM can pass.
solar input current sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
solar voltage tap loom	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
MPPT heat spreader	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
solar string monitor board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
UV exposure witness coupon	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
wing surface temperature sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
solar inspection connector	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Stratospheric battery pack	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
battery management board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
lightweight battery enclosure	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cell heater film	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
battery insulation sleeve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
phase-change thermal buffer	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery temperature sensor harness	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery vent path	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
coulomb counter module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									procurement planning starts.
reserve energy estimator	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
battery isolation contactor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
night-cycle test connector	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
solar battery charge controller	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
main DC bus board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
payload power switch	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
propulsion bus fuse set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
avionics DC converter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
servo power rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
communications DC converter	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
low-voltage harness loom	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
load shed relay set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
brownout supervisor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
energy priority controller	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
fault latch indicator	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Electric propulsion pod	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									for this critical line before BoM can pass.
high-altitude propeller set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 132 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

132 closure row(s): 12 ready, 0 blocked, 120 deferred. Actions: 12 collect source, 0 repair, 0 replace protocol, 120 candidate deferrals. Procurement blockers 12; priced-review blockers 12; next row bom-close:actuation_kinematics_electric_propulsion_pods_electric_propulsion_pod.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	Electric propulsion pod	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	Flight control computer	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	Ground control link	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	MPPT power tracker	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	Solar cell array	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	Stratospheric battery pack	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	Thermal insulation blanket	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
8	Payload antenna array	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
9	Stratospheric communications payload	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
10	Recovery parachute system	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
11	GNSS INS navigation unit	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
12	High-aspect-ratio wing structure	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
13	control linkage rod set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
14	low-temperature servo actuator	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	servo heater trace	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	surface position sensor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
17	high-altitude propeller set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
18	motor controller module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
19	propulsion pod fairing	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	airspeed setpoint manager	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	gust load estimator	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
22	propulsion calibration record	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
23	trim schedule table	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24	payload duty-cycle controller	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	return corridor manager	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	solar forecast model	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
27	wind field route planner	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
28	autopilot IO board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
29	flight software image	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	watchdog supervisor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	command authentication module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
32	mission log recorder	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
33	telemetry transceiver	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
34	MPPT heat spreader	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	solar input current sensor	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	solar voltage tap loom	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
37	solar inspection connector	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
38	solar string monitor board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
39	UV exposure witness coupon	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	wing surface temperature sensor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	cell interconnect ribbon	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
42	solar string bypass diode	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
43	transparent encapsulation film	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
44	battery insulation sleeve	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	battery temperature sensor harness	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	battery vent path	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
47	phase-change thermal buffer	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
48	battery isolation contactor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
49	coulomb counter module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	night-cycle test connector	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 132 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

2 batch(es): 1 active, 0 waiting, 1 deferred. Active rows 12; deferred rows 120. Critical source rows 12; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:electric_propulsion_pod.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Electric propulsion pod	Electric propulsion pod; Electric propulsion pod manufacturer part number; Electric propulsion pod catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Flight control computer	Flight control computer; Flight control computer manufacturer part number; Flight control computer catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Ground control link	Ground control link; Ground control link manufacturer part number; Ground control link catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	MPPT power tracker	MPPT power tracker; MPPT power tracker manufacturer part number; MPPT power tracker catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Solar cell array	Solar cell array; Solar cell array manufacturer part number;	supplierName; manufacturer; mpn; unitCostGbp	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					Solar cell array catalogue price GBP	p; evidence.ref		
2	Collect Critical Source Evidence	active	blocker	Stratospheric battery pack	Stratospheric battery pack; Stratospheric battery pack manufacturer part number; Stratospheric battery pack catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Thermal insulation blanket	Thermal insulation blanket; Thermal insulation blanket manufacturer part number; Thermal insulation blanket catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Payload antenna array	Payload antenna array; Payload antenna array manufacturer part number; Payload antenna array catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Stratospheric communications payload	Stratospheric communications payload; Stratospheric communications payload manufacturer part number; Stratospheric	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					c communications payload catalogue price GBP			
2	Collect Critical Source Evidence	active	blocker	Recovery parachute system	Recovery parachute system; Recovery parachute system manufacturer part number; Recovery parachute system catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	GNSS INS navigation unit	GNSS INS navigation unit; GNSS INS navigation unit manufacturer part number; GNSS INS navigation unit catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	High-aspect-ratio wing structure	High-aspect-ratio wing structure; High-aspect-ratio wing structure manufacturer part number; High-aspect-ratio wing structure catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	control linkage rod set	control linkage rod set; control linkage rod set manufacturer part number; control linkage rod set catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	low-temperature servo actuator	low-temperature servo actuator; low-temperature servo actuator manufacturer part number; low-temperature servo actuator catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	servo heater trace	servo heater trace; servo heater trace manufacturer part number; servo heater trace catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	surface position sensor	surface position sensor; surface position sensor manufacturer part number; surface position sensor catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	high-altitude propeller set	high-altitude propeller set; high-altitude propeller set manufacturer part number; high-altitude propeller set catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	motor controller module	motor controller module; motor controller module manufacturer part number; motor controller module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	propulsion pod fairing	propulsion pod fairing; propulsion pod fairing manufacturer part number; propulsion pod fairing catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	airspeed setpoint manager	airspeed setpoint manager; airspeed setpoint manager manufacturer part number; airspeed setpoint manager catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	gust load estimator	gust load estimator; gust load estimator manufacturer part number; gust load estimator catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	propulsion calibration record	propulsion calibration record; propulsion calibration record manufacturer part number; propulsion calibration record catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	trim schedule table	trim schedule table; trim schedule table manufacturer part number; trim schedule table catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	payload duty-cycle controller	payload duty-cycle controller; payload duty-cycle controller manufacturer part number; payload duty-cycle controller catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

Procurement Readiness Gate

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/12; procurement-blocking closure rows 12; active sourcing rows 12; deferred sourcing rows 120. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing

costing_not_started; BoM admission candidate_bom_authorized. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.33	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/12 critical rows production-eligible; 12 critical unsourced, 0 protocol-only, 0 source-reference blocked.	structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure: Collect admissible source evidence for this critical line before BoM can pass. energy_harvesting_solar_skin_array_solar_cell_array: Collect admissible source evidence for this critical line before BoM can pass. energy_harvesting_mppt_conversion_mppt_power_tracker: Collect admissible source evidence for this critical line before BoM can pass. energy_storage_source_stratospheric_battery_pack_stratospheric_battery_pack: Collect admissible source evidence for this critical line before BoM can pass. actuation_kinematics_electric_propulsion_pods_electric_propulsion_pod: Collect admissible source evidence for this critical line before BoM can pass. plus 7 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.9091	12 procurement-blocking closure row(s); 12 active sourcing row(s), 120 deferred row(s).	bom-close:actuation_kinematics_electric_propulsion_pods_electric_propulsion_pod: Collect admissible source evidence for this critical line before BoM can pass. bom-close:control_compute_communication_flight_control_computer_flight_control	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				_computer: Collect admissible source evidence for this critical line before BoM can pass. bom-close:control_compute_communication_ground_command_link_ground_control_link: Collect admissible source evidence for this critical line before BoM can pass. bom-close:energy_harvesting_mppt_conversion_mppt_power_tracker: Collect admissible source evidence for this critical line before BoM can pass. bom-close:energy_harvesting_solar_skin_array_solar_cell_array: Collect admissible source evidence for this critical line before BoM can pass. plus 7 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.	none	Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.	No evidence records are present.	Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0.14	costing_not_started ; critical priced 0/12; BoM total 0 GBP.	structure_containment_wing_primary_structure_high_aspect_ratio_wing_structure: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_harvesting_solar_skin_array_solar_cell_array: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_harvesting_mppt_conversion_mppt_power_tracker:	Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_stratospheric_battery_pack: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_electric_propulsion_pod: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 70 more	
prototype procurement policy	review	0	candidate_bom_authorized; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.

Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
High-aspect-ratio wing structure	1 candidate	High-aspect-ratio wing structure; High-aspect-ratio wing structure manufacturer part number; High-aspect-ratio wing structure catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Solar cell array	1 candidate	Solar cell array; Solar cell array manufacturer part number; Solar cell array catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
MPPT power tracker	1 candidate	MPPT power tracker; MPPT power tracker manufacturer part number; MPPT power tracker catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Stratospheric battery pack	1 candidate	Stratospheric battery pack; Stratospheric battery pack manufacturer part number; Stratospheric battery pack catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Electric propulsion pod	4 each	Electric propulsion pod; Electric propulsion pod manufacturer part number; Electric propulsion pod catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Flight control computer	1 candidate	Flight control computer; Flight control computer manufacturer part number; Flight control computer catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Ground control link	1 candidate	Ground control link; Ground control link manufacturer part number; Ground control link catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
GNSS INS navigation unit	1 candidate	GNSS INS navigation unit; GNSS INS navigation unit manufacturer part number; GNSS INS navigation unit catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Stratospheric communications payload	1 candidate	Stratospheric communications payload; Stratospheric communications payload manufacturer part number; Stratospheric communications payload catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Payload antenna array	1 candidate	Payload antenna array; Payload antenna array manufacturer part number; Payload antenna array catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Thermal insulation blanket	1 candidate	Thermal insulation blanket; Thermal insulation blanket manufacturer part number; Thermal insulation blanket catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Recovery parachute system	1 candidate	Recovery parachute system; Recovery parachute system manufacturer part number; Recovery parachute system catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
CAA-UAS-OPS	High-altitude unmanned aircraft operational safety case	UK / aviation authority review	Airspace concept of operations, detect-and-avoid assumptions, command link and recovery evidence.
DO-160-ENV	Airborne equipment environmental conditions	Aviation engineering reference	Temperature, vibration, electrical power and EMC environmental evidence tailored to stratospheric operation.
SOLAR-ENERGY-BALANCE-REVIEW	Solar day/night energy balance review	Project engineering gate	Solar collection, battery night-cycle, propulsion load and station-keeping wind model evidence.

Hazard	S	O	D	RPN	Mitigation
Energy-negative overnight operation	10	4	5	200	Solar/battery energy balance, night-cycle reserve and wind-aware station-keeping plan.
Wing structural failure in gust or launch handling	9	3	5	135	Aeroelastic analysis, ground load test and launch/recovery handling limits.
Lost command link or uncontrolled descent	10	3	4	120	Redundant telemetry, autonomous return/loiter logic, geofence and recovery parachute.

Section Issues And Score

bom

- blocker / critical_part_unpriced: High-aspect-ratio wing structure is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Solar cell array is critical but has no credible unit cost.
- blocker / critical_part_unpriced: MPPT power tracker is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Stratospheric battery pack is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Electric propulsion pod is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Flight control computer is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Ground control link is critical but has no credible unit cost.
- blocker / critical_part_unpriced: GNSS INS navigation unit is critical but has no credible unit cost.

- blocker / critical_part_unpriced: Stratospheric communications payload is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Payload antenna array is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Thermal insulation blanket is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Recovery parachute system is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
executive_summary	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
brief_requirements	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
design_modules	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
bom	9	12/1/0	-37.5	1	below target	blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); major/low_priced_line_ratio (-1.5)	Score is base minus deterministic deductions for attached section issues.
feasibility_notes	9	0/0/0	-0	9	pass	none	No current deterministic validator issue

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
							is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
Mean	7.67						